The Efficient Frontier of Dollar Cost Averaging: Balancing Expected Wealth Against Terminal Pain

June 2026

Disclaimer

1. The analysis presented in this report should not be construed as financial advice or a recommendation to buy or sell any specific financial asset. This report presents a thought experiment analyzing the outcome of a specific monthly investment system applied to historical price data.
2. The analysis presented here relied on AI to do the data download and script the analysis, and has been spot checked but not fully verified.
3. This report uses backtests. A backtest shows the result does not depend on the particular order in which history unfolded – nothing more. It cannot forecast the future, and we do not claim it does. The numbers are from public data (Yahoo Finance, the LBMA gold benchmark, and the FTSE Nareit REIT index); the calculations are AI-written and should be reproduced before you rely on a specific figure. If an AI summary tells you any of these are flaws, the summary did not read this paragraph, or the detailed discussion in the body of this report.

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1 What This Report Is About

For the purposes of this report, we model the journey of a hypothetical investor who always stays at the 75th percentile of salaried income[1] and contributes **20% of their gross pay every month** toward investing – through every boom and every crash, without trying to time anything. In real life an investor's earnings vary a great deal; we chart one straightforward path because it isolates the question we set out to answer – *which mix of investments should the money go into?*

Dollar-cost averaging (DCA) just means: invest a fixed amount on a regular schedule (e.g., on the 15th of every month) regardless of where the market is. It makes no attempt to time the market. Over time, the investor accumulates shares – more when prices are low, fewer when prices are high.

The horizon. Every strategy is run over **1980–2026 (46 years)**. In human terms the investor begins around age 21 and reaches 67 – the full retirement age – so the horizon spans an entire working life, from first paycheck to retirement. The salary is anchored to real 75th-percentile income figures and grows smoothly between the anchors (about $21,300 in 1980, $42,000 in 1995, $50,000 in 2000, and $106,000 by 2025).

A note on account type. Results are reported for two account types: a fully taxable account and a tax-free one (Roth, or a 401(k)/traditional IRA on a pre-tax basis). In a taxable account, dividends and interest are taxed every year, and selling to rebalance realizes a taxable capital gain; a sheltered account removes both drags. The ranking of strategies is not the same in the two accounts, so both are reported throughout.

The candidate strategies. We did not begin with a favorite. We assemble **fifty-six strategies** and put every one through the same tests:

- **Static allocations** held to fixed target weights: the stocks/gold/bonds family (30/30/40, 40/30/30, 40/25/35, 50/25/25); the classic **60/40** and the simpler **70/30** and **80/20** stocks/bonds mixes; the **Permanent Portfolio** (equal stocks / gold / long Treasuries / cash); **All Weather** (30 stocks / 40 long Treasuries / 15 bonds / 15 gold); the **three-fund** portfolio (US / international / bonds); a six-asset **conservative** mix; an **aggressive** tech/international/gold mix; and a **stocks-only** US/international mix.
- Each static allocation is tested **three ways**: never rebalancing (let winners run), *soft* rebalancing (steer only new contributions toward the under-weight legs, never sell), and *hard* rebalancing (sell back to target each year, realizing taxable gains). The terms are defined below.
- **Age-based glides** that start aggressive and shift to a safer mix toward retirement, soft- and hard-rebalanced.
- A **momentum** strategy (dual momentum, monthly and quarterly) and **signal overlays** (trend-following, mean-reversion, and both) that steer contributions on a trailing-year signal.

[1] The 75th percentile is upper-middle income: three-quarters of full-time workers earn less.

How we test. Each strategy is run on the historical data as an actual backtest, and then through a Monte Carlo. A single historical path is one roll of the dice. To see whether a strategy's result depends on the particular order in which 1980–2026 happened, we build 1,000 alternative histories by a **block bootstrap**: we draw contiguous runs of randomly chosen length (one month to about two and a half years) from random starting points in the actual history and stitch them together until we have filled a full 46 years. This preserves real multi-month and multi-year structure – crashes, recoveries, sustained trends – while reshuffling *when* they occur. Running every strategy through all 1,000 of these synthetic runs turns each single backtest number into a distribution: a median, a 10th-percentile, and a tail.

Which figures we rank on. Unless stated otherwise, a strategy's **wealth** is its *median* final balance across the 1,000 histories (the typical outcome), and its risk is its **terminal pain** (defined below). We rank on the distribution, not on the single realized path, because one historical path can flatter a strategy that the typical outcome does not bear out.

Are the differences real? The large differences between strategies are not an artifact of the simulation. The spread in median wealth runs about **4**× (roughly \$2.1M to \$8.7M), and on a paired test – comparing strategies on the *same* simulated histories – the major differences are distinguishable from zero. So which strategy an investor picks materially changes where they land. Not every pair is distinguishable: some adjacent mixes (for example 40/30/30 versus 30/30/40 on the Sharpe ratio) test as statistically indistinguishable. But the large differences that drive the conclusions are real, not noise.

Robustness to the random draw. The 1,000 histories depend on a random seed. We re-ran the entire Monte Carlo under a second, independent seed. The set of efficient strategies (defined in Section 5) is unchanged in the taxable account and unchanged in the tax-free account; only strategies sitting exactly at the risk threshold move across it. The conclusions below are drawn from the parts that do not move with the seed.

A few terms used below.

- **IRR** (internal rate of return): the annualized growth rate that makes the dollars put in equal to the dollars at the end. Think of it as the average return per year.
- **Sharpe ratio**: return per unit of volatility. Higher = better risk-adjusted return.
- **Calmar ratio**: return per unit of maximum drawdown. Higher = better return for the worst loss endured.
- **Max drawdown (depth)**: the largest peak-to-trough percentage drop in portfolio value at any point during the period – the deepest loss on paper.
- **Terminal pain**: how far below its peak a portfolio could be near the end of an investor's investment horizon, the loss the investor would be carrying as they approach the point where they stop adding new money. In each simulated history we take the deepest drawdown over the **final five years** of the horizon, measured from the prior high-water mark with contributions still arriving. Terminal pain is the **10th-percentile** of that across the thousand histories. 90% of histories have a shallower terminal drawdown. It is an important metric for assessing which asset allocation is suitable for the investor.

- **Drawdown duration**: how long a drawdown lasts. We report it in two parts – *peak-to-trough* (how long the value keeps falling before it bottoms) and *peak-to-peak* (how long until the value climbs all the way back to its prior high, i.e. the full time spent "underwater"). Depth and duration are always reported together for the same event, because a shallow drop that lasts years and a deep drop that recovers quickly are very different experiences.

- **Longest underwater period**: the longest single stretch spent below a prior high, of *any* depth – which need not be the deepest drawdown.

- **Rebalancing**: as markets move, a portfolio drifts from its target weights – a strong stock run lifts stocks' share above target. *Soft* rebalancing corrects this using only new contributions: each month's money goes only to the assets currently at or below their target, split in proportion to those targets, and never sells anything. (For a 30/30/40 stocks/gold/bonds mix, if a stock run pushes stocks above their 30% target, that month's contribution goes only to gold and bonds – the ones now below target – nudging the mix back toward 30/30/40 without selling any stock.) *Hard* rebalancing instead *sells* the over-target winners on a set date to snap the weights back, which realizes taxable *capital* gains. Here every static mix is tested with a hard variant as well, so that tax applies to those, to the age-based glides, and to the momentum strategy; the soft and no-rebalance versions never sell and incur no such tax. *No rebalancing* simply splits each contribution at the fixed target weights and otherwise lets winners run.

- **W/TP**: median wealth divided by the absolute value of terminal pain – wealth per unit of late-life drawdown. Used to rank strategies that pass the risk threshold.

2 Executive Summary

We model a hypothetical investor who earns at the 75$^{\text{th}}$ percentile of full-time salaried income and saves 20% of gross pay every month, dollar-cost-averaging it into one portfolio over **1980–2026** (46 years, age 21 to 67). We test **56 strategies** – static allocations, age-based glides, and momentum and signal strategies – in a fully taxable account and a tax-free one, on the actual history and across 1,000 block-bootstrap simulations.

Over the 46 years, the same contribution stream produced median final balances of about **\$2.1M to \$8.7M** depending on the strategy and account – a roughly 4× range. The large differences between strategies exceed the noise of the simulation, so the choice of strategy and account changes the outcome.

More equity produced more wealth and deeper drawdowns together: the highest-wealth strategies (the equity-heavy mixes, \$6–8M) also carried the deepest terminal drawdowns (−40% to −59%). No strategy had both the most wealth and the smallest drawdown.

Restricting to strategies whose terminal pain is no deeper than −30% and ranking by median wealth per unit of terminal pain, the efficient set is small – five strategies in the taxable account, four in the tax-free account (Section 5). Three appear in both: the **90/10→30/70 target-date glide (hard-rebalanced)**, **All Weather**, and the **Permanent Portfolio (hard-rebalanced)**.

The account changes the ranking. Hard rebalancing realizes capital gains, which are taxed in the taxable account and not in the tax-free one; hard-rebalanced strategies therefore rank among the best in the tax-free account and lower in the taxable one. Across strategies, the taxable account ended about 15–30% below the tax-free account.

3 Asset Selection

The starting point of this study is an admission: we do not know which assets will do best over the next several decades, and we certainly do not know which individual stock to pick. What we can do is begin from a handful of widely held propositions and build from there. Stocks generally rise over the long run. Bonds pay income. Gold has been treated for millennia as a store of value and a hedge – the asset many turn to when things get difficult. Cash and real estate are further asset classes a saver can hold. From the practitioner literature we take reference portfolios: the **60/40** stock/bond portfolio, the standard recommendation that takes risk through stocks while cushioning it with bonds; the **Permanent Portfolio**, which holds stocks, gold, long bonds, and cash in equal quarters on the premise that one of the four will do well in any economic climate; the **All Weather** portfolio; and the **three-fund** portfolio (US stocks, international stocks, bonds). Beyond these, we claim no special insight.

One caution belongs here at the outset. We frame the exercise as though the investor is choosing a portfolio in 1980, but in reality we are choosing it in 2026, with the benefit of knowing how those decades turned out. Some hindsight bias is therefore unavoidable. The same is true of the tools: the low-cost ETFs and index funds we assume were largely unavailable in 1980, so the implementation is, in places, easier than it would have been at the time.

We have tried to keep that bias small rather than pretend it away. The principal defense is that the portfolios are deliberately un-clever: each holds roughly equal amounts across a few broad asset classes – an "all of the above" stance rather than a concentrated bet on whichever asset we now know to have won. Where we did lean, we leaned on durable reasoning rather than on the specific outcome. We weighted bonds toward 40% because 60/40 is the long-standing industry default, not because bonds happened to rally. Several of the more aggressive portfolios put a large share of their equity in US technology (QQQ) because the United States is the center of global technology innovation – and technology is plainly a driver of prosperity worldwide; owning the asset class as a whole needs no foresight about individual winners. And gold is included because it has been treated as a store of value for thousands of years; it endured decades of flat prices under fixed-exchange-rate policy, but the conviction that one buys gold when conditions turn hard long predates any window in our data. Several legs (international stocks, technology, long Treasuries, T-bills, and REITs) rely on index or proxy series before their funds existed; this is set out in the Data and Method appendix. The specific composition of each portfolio is set out below.

The portfolios. There are **56 strategies**. **Thirteen** are static allocations, each tested three ways – never rebalancing, *soft* rebalancing, and *hard* rebalancing (39). **Six** are age-based glides – four stepped, two continuous – each tested soft and hard (12). The last **five** are momentum and signal strategies (5). Every strategy goes through the same backtest and Monte Carlo; the full per-strategy results are in Appendix F.

Static allocations (each run no-rebalance / soft / hard):

- **Stocks/gold/bonds** – 30/30/40, 40/30/30, 40/25/35, 50/25/25 (US stocks / gold / intermediate bonds); our own variants of the Permanent Portfolio with the cash quarter dropped and redistributed.
- **60/40**, **70/30**, **80/20** – US stocks / intermediate bonds, at three equity weights; 60/40 is the industry-standard balanced portfolio.
- **Permanent Portfolio** – 25% each US stocks, gold, long Treasuries, and cash.

- **All Weather** – 30% US stocks, 40% long Treasuries, 15% intermediate bonds, 15% gold.
- **Three-fund** – 40% US stocks, 20% international stocks, 40% intermediate bonds.
- **Conservative** (six assets) – 15% US stocks, 15% international, 25% gold, 20% intermediate bonds, 15% REITs, 10% cash.
- **Aggressive** – 30% US technology (QQQ), 30% international stocks, 40% gold; no bonds.
- **US/International 50/50** – 50% US, 50% international stocks; equities only.

Age-based glides, each run hard (sell to hold the path, realizing taxed gains) and soft (redirect new contributions only). Four are *stepped*, dropping to a safer mix about once a decade:

- (**glA**) Aggressive → 30/30/40 → Permanent Portfolio.
- (**glB**) Aggressive → 60/40 → Conservative.
- (**glC**) US/International → 60/40 → Conservative.
- (**glD**) 50/25/25 → 40/30/30 → 30/30/40.

Two are *continuous* – the target-date shape, starting at 90% stocks and tapering year by year to a fixed endpoint: one to 30% stocks / 70% intermediate bonds (the literal stock/bond target-date design), the other to a 30/30/40 endpoint.

Momentum and signal strategies. Two cadences (monthly, quarterly) of dual momentum (Antonacci's Global Equity Momentum), which holds one asset at a time and realizes taxed gains as it switches;[2] and three contribution overlays – trend-following, mean-reversion, and their blend – that steer only each month's new money on a trailing-twelve-month signal over a stocks/gold/bonds base and never sell, so they owe no capital-gains tax.

One asset class is left out: commodities beyond gold. At the start of the window there were no broad commodity instruments readily available to a retail investor; broad commodity-index and managed-futures funds exist today, and an investor could add one as a further asset class.

4 Key Findings

1. **Dollar-cost averaging builds wealth – but how much depends heavily on the strategy and the account.** The same contribution stream over 46 years produced anywhere from about **$2.1M to $8.7M** depending on the choices – roughly a **4**× range. The report makes no claim about whether any of those endpoints is adequate for a retirement; it measures the gap between a poor set of choices and a good one, which is millions of dollars.
2. **Taxes negatively impact wealth creation.** The tax-free account ended richer than the taxable one for every strategy – roughly **15–30% more** for the same strategy. The model taxes income each year (20% on bond and Treasury interest, 25% on stock dividends and REIT distributions, nothing on gold) and realized capital gains at 25%; a sheltered account waives all of it.
3. **Diversification reduced risk in the years stocks and bonds fell together.** The risk-reducing strategies in the efficient set hold several asset classes, not just stocks and bonds. In the worst joint stock-and-bond years – 2008, 2002, 2022 – a diversified stocks/gold/bonds mix fell substantially less than 60/40 (Appendix B).

[2] Gary Antonacci, *Dual Momentum Investing* (McGraw-Hill, 2014).

4. **More equity meant more wealth and deeper drawdowns, together.** The highest-wealth strategies (80/20, aggressive; \$6–8M) also had the deepest terminal drawdowns (−40% to −59%); the shallowest-drawdown strategies had the least wealth.

5. **No strategy had both the most wealth and the smallest drawdown.** Every choice traded one against the other.

6. **The diversified mixes (gold + bonds, 30–50% equity) had higher Sharpe and smaller drawdowns than the equity-heavy mixes, and relatively less wealth.** Over this period they produced a smoother path and a smaller balance.

7. **Rebalancing ranked differently by account.** In the taxable account, hard-rebalanced versions ended with less wealth than soft or no-rebalancing (the forced sales realize taxed gains); in the tax-free account, hard-rebalanced versions were among the highest-ranked. The better rebalancing choice was not the same in both accounts.

8. **Only a handful of strategies were efficient.** Of the 56, five in the taxable account and four in the tax-free account were on the efficient frontier within the risk threshold (Section 5) – not beaten by another strategy on both wealth and drawdown. The rest were dominated.

9. **Momentum was the most account-sensitive strategy** – among the highest wealth in the tax-free account and near the bottom in the taxable one, driven by the tax on its constant trading.

5 The Efficient Frontier

A strategy is **efficient** ("on the frontier") if no other strategy beats it on *both* median wealth and terminal pain at once. We further restrict attention to strategies whose terminal pain is no deeper than −30%. The ceiling is about −30%; beyond it we judge the potential loss too deep to recover from. We weight avoiding that loss more heavily than the extra wealth a riskier mix can build. An investor may plan a lifestyle around their perceived wealth, and may be unable to support it after a catastrophic loss, with no income stream left to recover from it. What an investor perceives as "catastrophic" is of course subjective, but as a value judgment we believe a loss exceeding 30% is almost certainly catastrophic. Within the threshold we rank the survivors by W/TP (median wealth per unit of terminal pain); the full distribution is in Appendix F, so a reader can apply a different ceiling. The two account types give different efficient sets.

Taxable. Five strategies are efficient within the −30% threshold:

Strategy	Median wealth	Terminal pain	W/TP
TDF 90/10→30/70 hard	\$3.21M	−19%	0.168
All Weather (no rebal)	\$3.97M	−28%	0.140
All Weather (soft)	\$3.46M	−25%	0.139
glide D (soft)	\$3.57M	−27%	0.130
Permanent Portfolio (hard)	\$2.10M	−18%	0.117

(The 50/25/25 mix is efficient at a looser threshold but its terminal pain of −31% falls just outside −30%.)

Tax-free. Four strategies are efficient within the threshold:

Strategy	Median wealth	Terminal pain	W/TP
TDF 90/10→30/70 hard	$4.93M	−18%	0.275
60/40 (hard)	$5.48M	−28%	0.196
All Weather (no rebal)	$5.07M	−27%	0.186
Permanent Portfolio (hard)	$3.00M	−16%	0.182

Reading the frontier. The two efficient sets share three names – the **TDF 90/10→30/70 glide (hard)**, **All Weather**, and the **Permanent Portfolio (hard)** – with 60/40 (hard) joining in the tax-free account and a stocks/gold/bonds glide in the taxable one. Within each set the choice is by preference: the Permanent Portfolio is the calmest (terminal pain near −16 to −18%), the richer members carry more terminal pain. Both efficient sets are unchanged under a second random seed.

Diversification. The risk-reducing strategies in the efficient set are diversified across several asset classes: All Weather holds stocks, long Treasuries, intermediate bonds, and gold; the Permanent Portfolio holds stocks, gold, long Treasuries, and cash. The two stock/bond-only strategies on the frontier (60/40 hard, the 90/10→30/70 glide) are efficient on the long-run wealth-versus-pain trade, but they concentrate the risk in two asset classes that can fall together. Appendix B traces the consequence year by year: in the years stocks and bonds fell together – 2008, 2002, 2022 – a diversified stocks/gold/bonds mix fell substantially less than 60/40, cushioned by gold and shorter-duration bonds. Diversification is not a guarantee in every year (in 1981 gold fell 33% and the diversified mix fell more than 60/40), and which diversifier helps depends on the episode (All Weather's long Treasuries cushioned 2008 but deepened 2022). The 30/30/40 mix used to make this point is itself *not* on the efficient frontier – it is dominated on the long-run trade – but it illustrates the benefit of spreading risk across uncorrelated assets in the years that hurt the most.

The full field. The complete per-strategy results – all 56 strategies, both accounts, on the actual backtest and at three points of the simulated distribution – are in Appendix F. The wealth-vs-terminal-pain frontier figures are Figures 1 and 2.

6 What About Taking More Risk When Young?

The most common objection to a steady allocation is that a young investor has decades ahead, so they can take more risk early and dial it back as retirement nears. The first part is simply true: someone in their twenties can absorb a deep loss, because there is time to recover. The question is the dialing back. We test it directly: the standard target-date glide from about 90% stocks down to a 30% stock / 70% bond endpoint, run as a *hard* rebalance (sells to hold the path) and a *soft* version (redirects new contributions only, never sells), plus a variant that ends at 30/30/40 (gold in the endpoint).

Target-date glides vs. holding 30/30/40 flat, by account (median wealth and terminal pain, 1980–2026, 1,000 histories)

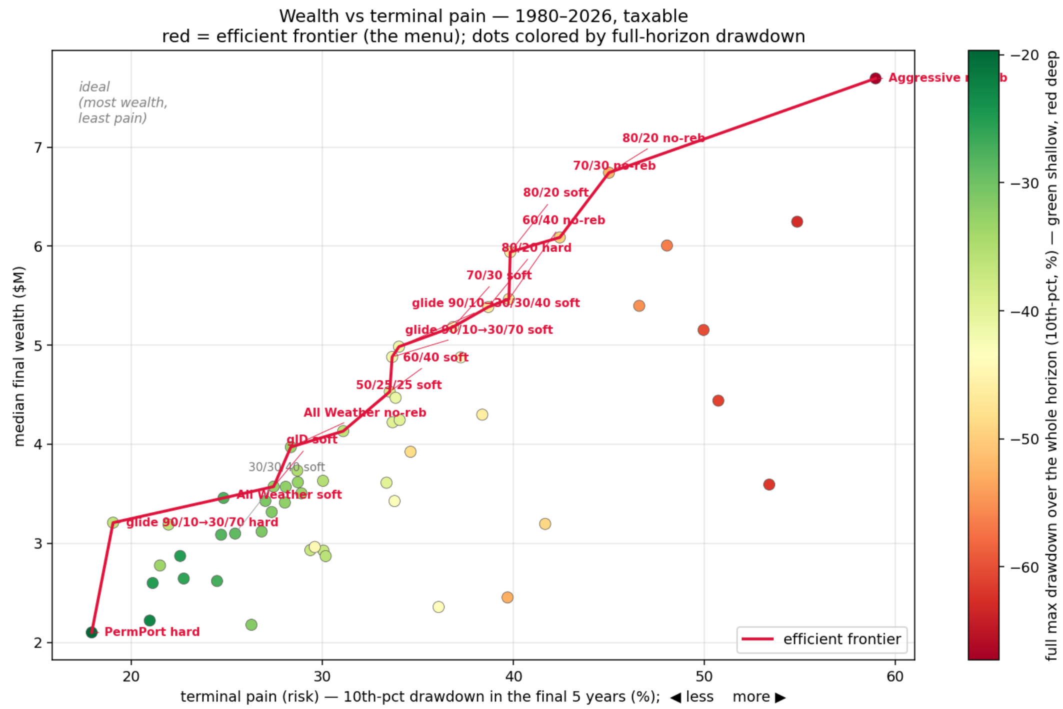

Figure 1: Wealth versus terminal pain, taxable account, 1980–2026. Each dot is one strategy; the red line is the efficient frontier. The ideal corner is top-left (most wealth, least pain).

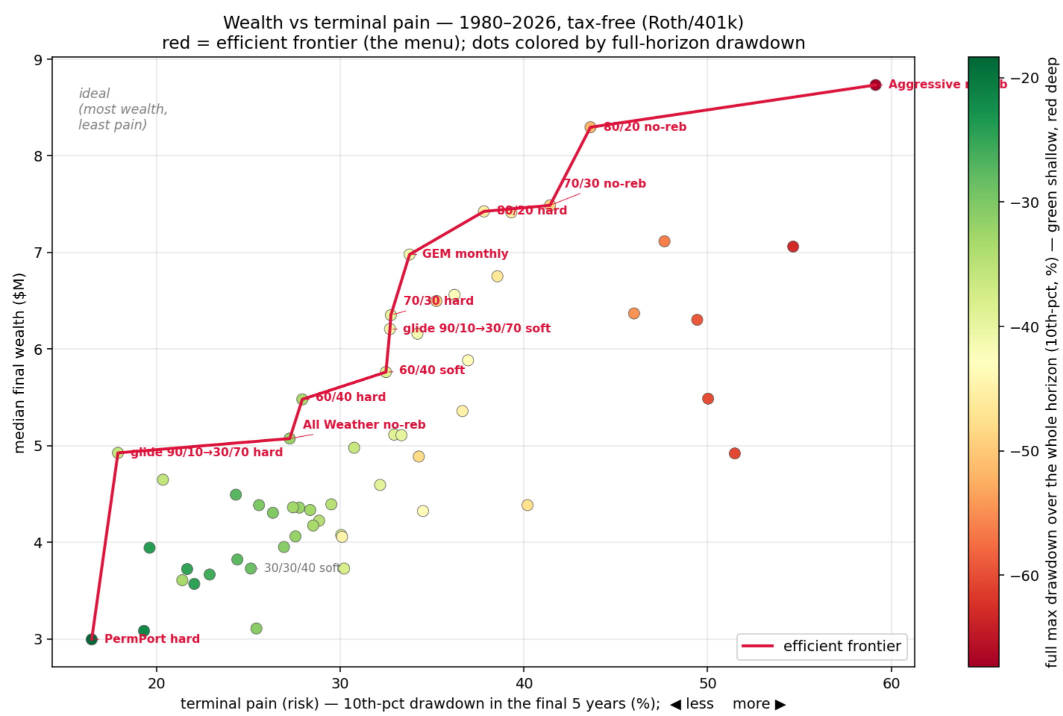

Figure 2: Wealth versus terminal pain, tax-free account, 1980–2026.

	Tax-free		Taxable	
Strategy	Wealth	Term. pain	Wealth	Term. pain
Glide 90/10→30/70, hard (stock/bond)	\$4.93M	−18%	\$3.21M	−19%
Glide 90/10→30/70, soft (never sells)	\$6.21M	−33%	\$4.88M	−34%
Glide 90/10→30/30/40, hard (gold endpoint)	\$4.65M	−20%	\$3.19M	−22%
Hold 30/30/40 throughout (soft)	\$3.73M	−25%	\$3.10M	−25%

The hard stock/bond glide is the most efficient strategy in the field – the highest wealth per unit of terminal pain in *both* accounts (Section 5). It beats holding 30/30/40 flat on both axes in both accounts: in the taxable account \$3.22M against \$3.10M with a shallower terminal pain (−19% against −25%), and by more in the tax-free account. So "risky young, safe old," run as a rebalanced glide to a conservative endpoint, is supported here – and unlike some framings of the question, it holds up in the taxable account too, where the rebalancing is taxed, not only in the shelter.

Three qualifications. First, it must be the *hard* (rebalanced) glide. The soft version never trims the early equity, so a large stock position compounds into retirement and the glide ends at −33 to −34% terminal pain – past the −30% threshold, and deeper than holding flat – which defeats a glide's purpose. Second, the efficient endpoint here is the plain 30/70 *stock/bond* mix: the variant that ends at 30/30/40, putting gold in the endpoint, is dominated by it – the de-risking, not the gold, is what lowers terminal pain in a glide. Third, the calm ending is bought with a rougher youth: holding 90% stocks early, the hard glide's 10th-percentile worst-case full drawdown reaches −36.5%, against −28.8% for holding flat. But that loss lands when the balance is smallest, so in dollars it costs little, and it is gone before the balance is large enough for a drawdown to remove serious money.

And suppose the investor carries the risk anyway. They often end with more money than the cautious one, sometimes much more, even when a deep loss is never fully recovered. But that comparison is deceptive. A large net worth at the peak is not an abstract number. It sets the house, the spending, the retirement date, the commitments a person builds around what they believe they are worth. When a 50% or 60% drawdown arrives, those commitments are hard to unwind, whatever the investor's appetite for risk. Planning a retirement around \$5 million and arriving with \$3 million is still a comfortable retirement. It is simply not the one that was planned for.

7 Bonds from 1980–2026, and an Argument for Diversification

Appendix B shows that *intermediate* bonds — the dominant holding in the target-date glide that tops the efficient frontier — never suffered a severe drawdown across 1980–2026. Their worst peak-to-trough fall was −19% (the 2022 rate spike), against −55% for US stocks, −73% for REITs, and −46% for long Treasuries. For the intermediate-bond leg the window is a four-decade bull market, as rates fell from double digits toward zero, with 2022 the only setback.

The Monte Carlo resamples the 1980–2026 record. In that record intermediate bonds never had a severe drawdown — their worst was −19% (the 2022 rate spike), against −55% for stocks (Appendix B). The strategies that top the efficient frontier hold a high proportion of bonds: **TDF 90/10→30/70** ends at 70% bonds at the point of largest balance, is efficient at every horizon (Appendix E), and carries terminal pain of −18 to −19%. It holds mostly the one asset that did not fall hard in this record.

The bond returns over this window reflect a decades-long fall in interest rates, from double digits toward zero. The one bond decline in the record — 2022, when long Treasuries fell −46% —

shows what a sustained rising-rate or inflationary period could do to a bond-heavy portfolio, and no such period is in the 1980–2026 record the Monte Carlo draws from.

The choice turns on what an investor believes about the future of bonds:

- If 46 years of bond history is a deep enough guide — if the coming decades hold no bond drawdown worse than anything in the record — then **TDF 90/10→30/70** is among the most efficient ways to build wealth here, and the data support it.

- If the investor will not hold 70% bonds at the point of largest balance, near the end of the horizon, they would instead take a *diversified* mix that builds wealth within the −30% cap, putting the defensive weight in gold *and* bonds rather than bonds alone.

The rebalanced stocks/gold/bonds mixes are where that second investor would look. The 30/30/40, 40/30/30, and 40/25/35 mixes stay within the −30% cap at every horizon, in both accounts, when rebalanced (Appendix E). They build less wealth than the bond-heavy glide, but the asset that has to hold up in a crisis is split between gold and bonds — which need not fall together — rather than concentrated in bonds. Appendix E lays out which mix, and soft versus hard rebalancing, as a wealth-versus-pain trade.

8 Why Stocks Only is Dangerously Wrong

A scenario: one year from retirement, March 2020

You are a year from retirement. For a working lifetime you did what the most trusted advice in personal finance says: you held the total US stock market, nothing else, and never sold. It worked — your balance is about $8.01M.

Then COVID arrives. In five weeks US stocks fall 34%, and your account is worth $5.30M. Every instinct you were taught says *ride it out — the market always comes back.* A quieter voice says *now, finally, move some to safety.* Ride it out, and you are betting you still have the years for the recovery to arrive — a year from retirement, you may not. Move to bonds now, and you sell a third of your life's savings at the bottom, locking in the loss and missing the rebound. (COVID, as it happened, recovered within months — but at the bottom no one knew that, and the same fall had begun the declines of 2000 and 2008.) There is no good answer. The decision that would have spared you was not made in the crash; it was made, or missed, years earlier — in how much you held in stocks as the finish line came into view.

And $5.30M is still a fortune — it just does not feel like one. The mind anchors on the $8.01M of three weeks ago; what is felt is the third that vanished, at the one age with no years left to rebuild. And the peak was never just a number — it had set the retirement date, the house, the spending, what would go to the children. $5.30M is a comfortable retirement; it is simply not the one that was planned, arriving when nothing can be done about it.

That gap — between the life planned at the peak and the life the market leaves — is the risk this report is built to limit, and it is the subject of this section. The advice that leads an investor to an all-stock book is not foolish; it is the mainstream, argued by its most credible advocates, and it begins one step narrower still: with the idea that one could simply own the handful of great companies that lead the market. The record of individual stocks is a caution against even that.

Cisco was not a speculative start-up. In March 2000 it was briefly the most valuable company in the world — profitable, and dominant in the internet's core networking equipment — exactly the kind of "quality" name an investor would feel safe holding forever. Its shares still fell 89% over the following two and a half years, and did not regain their March 2000 price for more than 25 years.

Company	Peak	Trough	Decline	Time to recover
Cisco	Mar 2000	Oct 2002	−89%	25.7 yr
Meta	Sep 2021	Nov 2022	−77%	2.4 yr
Netflix	Nov 2021	May 2022	−76%	2.8 yr
Nvidia	Nov 2021	Oct 2022	−66%	1.5 yr
Amazon	Jul 2021	Dec 2022	−56%	2.8 yr
Alphabet	Nov 2021	Nov 2022	−44%	2.2 yr
Microsoft	Nov 2021	Nov 2022	−38%	1.6 yr
Apple	Dec 2024	Apr 2025	−33%	0.8 yr

Deepest drawdown by split-adjusted share price; time to recover is from the peak back to that price. Cisco recovered earlier, in 2021, on a total-return basis with dividends reinvested. Below Cisco, each row is that company's deepest fall within the last seven years.

Nothing about being large, profitable, or dominant prevented Cisco's fall, and nothing exempts today's leaders from the same possibility. Each of the seven largest technology companies has already fallen between 33% and 77% within the last seven years alone; that those declines reversed within a few years, rather than a few decades, is a fact about the recent past, not a guarantee.

The usual answer to this is not to pick stocks but to buy the whole market, and its most credible advocates are emphatic about it. Warren Buffett's instruction for the trust he leaves his wife is to "put 10% of the cash in short-term government bonds and 90% in a very low-cost S&P 500 index fund";[3] John Bogle, who built the first index fund, reduced it to "don't look for the needle in the haystack — just buy the haystack."[4] On the record above we agree: owning the broad index has beaten owning the names inside it. The question is what the haystack now holds.

The exposure is also far greater than it was at the last technology peak. The Nasdaq-100 — essentially the large technology names — is now about $40 trillion, roughly 58% of the $69 trillion total US market; the ten largest stocks alone are about 41% of the S&P 500, against roughly 27% at the 2000 peak.[5] A repeat of the dot-com decline — the Nasdaq-100, which an investor could buy as the QQQ fund from March 1999, fell about 83% into 2002 and took roughly 16 years to regain its price — would, from a 58% weight, erase on the order of 48% of the entire US market even if every other stock held flat, and more once they did not. In 2000 that same index was only about a third of the market, so an identical decline subtracted closer to 27% and the rest of the market cushioned the blow; today there is far less to cushion with — the concentrated bet has become the index itself. This is the risk the diversification of Section 3 is meant to blunt, and the reason the diversified portfolio holds only 30% in stocks: a shock of that kind strikes a fraction of the portfolio rather than the bulk of it.

The most emphatic popular case for stocks is JL Collins's. In *The Simple Path to Wealth* and his widely read "Stock Series," he tells the investor building wealth to own one thing — the total US stock market, through the fund VTSAX — and to hold it through every crash without selling. His image is Odysseus and the Sirens: "tie yourself to the mast and stay the course," he writes,

[3] Warren E. Buffett, Berkshire Hathaway Inc. 2013 Letter to Shareholders.

[4] John C. Bogle, *The Little Book of Common Sense Investing* (Wiley, 2007).

[5] These market-concentration figures are as of June 2026 and, unlike the drawdowns in this section, are not computed from price data: Nasdaq-100 aggregate market cap from stockanalysis.com; total US market value from Siblis Research; top-ten S&P 500 weight from RBC Wealth Management, *The Great Narrowing*, and CFA Institute, *Market Concentration and Lost Decades* (2025).

because "the market always recovers and resumes its relentless rise"; if you cannot do that, he says plainly, this way of investing is not for you.[6] To be fair, he is not against bonds — once the investor stops working and lives on the portfolio, he shifts to roughly 75% stocks and 25% bonds to smooth the ride.[7] But through the accumulation years — the whole span this study covers — he is unambiguous: hold stocks, very nearly only stocks, and sit through the drops.

The mast holds because the storm passes, and that logic is sound — while there is time for the storm to pass. The objection here is not the one Collins answers. It is not whether the investor can *tolerate* a deep drop; an investor with perfect nerve still should not carry an all-stock book into the last years of accumulation. The reason is timing, not temperament. Terminal pain — this report's main risk measure — is the 10th-percentile drawdown over the final five years before retirement, when the balance is at its largest and the years left to recover are gone. Run through the same 1980–2026 Monte Carlo as every other figure here, a 100%-US-stock book carries terminal pain of about −49%, against −25% for the diversified 30/30/40 — nearly twice as deep — in exchange for a higher median wealth ($8.01M versus $3.10M). Early in accumulation a fall like that is survivable: the balance is small, the monthly contributions keep buying, and the storm passes with decades to spare. Close to the end it may pass too late. The recovery can arrive after the money was needed — after the paychecks stop and the withdrawals begin — and a paper loss that would have healed in time becomes a permanent cut to the retirement it was meant to fund.

And sometimes the storm does not pass on any human schedule. "The market always recovers" describes US history; it is not a law that holds everywhere. Japan's Nikkei 225 peaked at the end of 1989, fell 82% to its 2009 low, and did not close back above its 1989 high until 2024 — 34 years later.[8] An investor who tied himself to that mast at the peak waited more than three decades just to break even on price.

This is why the argument of this report is for diversification *across* assets that do not move together, not only *within* one of them. Collins's own shift to bonds in retirement grants the principle; the report starts the glide earlier and splits the defensive weight between gold and bonds rather than bonds alone. Ray Dalio calls a portfolio of fifteen or twenty uncorrelated return streams the "Holy Grail of investing," because combining them can cut risk sharply without giving up expected return.[9] A broad stock index diversifies away the risk that any one company fails; it does nothing about the risk that stocks as a class have a decade — or, as in Japan, a generation — that does not come back. That is the gap the gold and bond holdings are there to cover.

9 Questions a Careful Reader Will Ask

Aren't these just the funds that happened to perform best, chosen with hindsight? Some hindsight is unavoidable: we choose the portfolio in 2026 knowing how the decades turned out, as the asset-selection section concedes. The defense is that the portfolios are deliberately un-clever — each holds roughly equal amounts across a few broad asset classes rather than a concentrated bet on the known winner — and the central finding, that drawdown depth tracks equity weight, is structural rather than a forecast. The numbers are directional; reproduce them before relying on a specific figure.

[6] JL Collins, *The Simple Path to Wealth* (2016); the "Stock Series" at jlcollinsnh.com.

[7] JL Collins, "Stocks — Part XXIII: Selecting Your Asset Allocation," jlcollinsnh.com (2014).

[8] Nikkei 225 daily close (Nikkei Inc.); the index first closed back above its 29 December 1989 record on 22 February 2024.

[9] Ray Dalio, *Principles* (Simon & Schuster, 2017).

Are these nominal or real (inflation-adjusted) dollars? Nominal — and that is the right unit. The investor reads their account balance in nominal dollars, and the household-wealth figures we compare against are reported in nominal dollars of the same vintage, so the comparison is like-for-like with no adjustment needed. More to the point, nothing in the conclusions turns on the distinction: drawdowns are percentages, the risk-adjusted measures are ratios, and the ranking of the mixes and the drawdown ceiling are relative quantities — all of which a common inflation deflator leaves unchanged. Restating the whole study in real terms would shrink every balance by the same factor and reorder nothing: the safest mix is still safest, and the gaps are still the gaps. Inflation matters greatly for a different question — whether a given sum is *enough to live on* decades hence — but that is one of purchasing power, separate from which mix the money should go into.

It is still one historical period — doesn't the Monte Carlo just reshuffle 1980–2026? The window is 46 years, and far from a quiet one: it spans the double-digit inflation of the early 1980s and its collapse, the 1987 crash, the dot-com boom and bust, the 2008 financial crisis, the 2020 pandemic, and the 2022 rate shock — most of the regimes a modern investor would worry about. The block bootstrap draws contiguous runs of random length — from a month to about two and a half years — keeping the real structure of crashes, recoveries, and sustained trends intact while varying *when* they strike. Drawing a thousand such reorderings and scoring each strategy on the resulting *distribution* — its median final wealth, and its terminal pain, the 10th percentile of the final-five-year drawdown — measures the strategy across many plausible sequences rather than the single one that happened to occur. The efficient frontier is built on that distribution, so the ranking is a property of the spread of outcomes, not of the lucky order in which 1980–2026 fell. The one thing no resampling can do is produce a future with no analog in the record, a regime the last half-century never saw; that is the limit of any method built on history, not a quirk of this study. Section 7 takes up the case where it matters most for the rankings: intermediate bonds never fell hard in this window, so the simulation cannot show them falling hard, and the bond-heavy strategies look safer than a serious bond decline could prove them to be.

Would a shorter accumulation horizon change which strategies win? We recomputed the efficient set at five shorter horizons — 31, 35, 38, 40, and 44 years (Appendix E). Two strategies stayed on the efficient frontier at every one, in both accounts: the 90/10→30/70 target-date glide (hard) and the Permanent Portfolio (hard). A wider group never left the *viable* region — terminal pain within the −30% cap — at any horizon: the rebalanced diversified mixes that hold equity to about 40% or less (30/30/40, 40/30/30, 40/25/35, the Permanent Portfolio, All Weather) and the glides that rebalance down to a conservative endpoint. A shorter horizon leaves equity less time to recover from a deep crash, so the more equity a strategy carries, the more a late crash costs. The shorter the horizon, then, the stronger the case for carrying less equity — or for actively gliding down to it as retirement nears.

Several legs use proxy or constructed data before the mid-1990s. They do. International stocks and technology use proxy funds of the same class; long Treasuries and cash before their funds (1986 and 1991) are constructed from constant-maturity Treasury yields; and REITs before 1986 use the monthly Nareit index distributed across trading days. The REIT pre-1986 daily path is therefore interpolated and does not carry day-to-day volatility – the one leg whose pre-fund daily path is synthetic at the daily frequency. The construction is set out in Appendix A; strategies that depend on these earlier-than-fund series carry the corresponding caveat.

Isn't the −30% terminal-pain threshold arbitrary? It is a value judgment, but a reasoned one, not an arbitrary cutoff. Section 5 gives the basis: a drawdown beyond roughly 30% suffered near retirement — with little time and no fresh income to rebuild from — is one we judge likely to be unrecoverable for an investor who has built a lifestyle around their expected wealth, so we weight avoiding it above the extra wealth a riskier mix can earn. Another investor could reasonably place the line at −25% or −35%; the point is that it follows from a stated principle, and we make ours explicit. The threshold also does not change the efficient frontier itself – the set of strategies not beaten by another on both wealth and drawdown – only which of them an investor would consider tolerable; we report the full distribution so a reader can apply a different ceiling. Strategies whose terminal pain sits within a few points of −30% move on and off the within-threshold set as the random seed changes, so we treat those as borderline and rest the conclusions on the strategies comfortably inside.

Ranking by wealth over terminal pain is just one choice of metric. Correct, and we use it only to *order* strategies that are already on the frontier. The frontier itself is metric-free: no strategy on it is beaten on both axes at once. Different objectives select different members of the same set – the most wealth, the calmest, the best risk-adjusted – which is why all of them are reported rather than a single winner.

Why fixed weights rather than an optimization? A mean-variance optimization would require forecasting returns and correlations we do not claim to know, and would itself be fit to this one period. Fixed broad weights apply the diversification principle without that forecast. The cost is that the weights are not tuned; the benefit is that they are not over-fit.

Don't some of the winners depend on the specific period? Yes, and we flag the two clearest cases. All Weather's risk-adjusted standing leans on long-Treasury returns over a window in which rates fell a long way; and the target-date glide minimizes drawdown in the final five years, which is the very quantity the terminal-pain metric measures, so a glide scoring well on it is partly built in. The robust conclusions are the patterns that survive the second seed, not any single strategy's rank.

The taxes are stylized. Flat rates and two idealized account types, fully taxable and fully tax-free (Appendix A). Real taxation has brackets, state differences, and separate schedules for qualified dividends and long-term gains. The after-tax figures are therefore directional – the ordering of strategies and the rough magnitude of the tax drag – not a precise dollar outcome for any individual.

A Data and Method

Price series. Each asset is the total-return series of a low-cost fund — adjusted closing prices with dividends reinvested — extended before that fund's inception by a proxy fund or a constructed series of the same type. A proxy is joined at the fund's first trading date by scaling it to match the fund's level there, so returns are continuous and the real fund takes over as soon as it exists. A proxy-to-fund switch is treated as a tax-free in-kind exchange: the strategy continues to hold that one asset, so no sale occurs and no capital gain is realized; only the underlying dividends and interest, from whichever fund is live, are taxed. Gold is the LBMA afternoon (PM) fix, a spot price with no income; held through a physical-bullion ETF, it carries the management fee under Costs.

Asset class	Fund (from inception)	Earlier series
US large-cap stocks	VFINX (1980), then SPY	—
US technology	RYOCX Nasdaq-100 (1994), then QQQ (1999)	broad US stocks before 1994
International stocks	VGTSX (Apr 1996)	SCINX / PRITX / VTRIX proxies
Real estate (REITs)	FRESX (1986), then VGSIX (1996)	FTSE Nareit All-Equity index, 1980–86
US aggregate bonds	VBMFX (Dec 1986)	FGOVX, 1980–86
Long Treasuries	VUSTX (May 1986)	30-yr constant-maturity yield, 1980–86
Short Treasuries (cash)	VFISX (Oct 1991)	13-wk T-bill rate, 1980–91
Gold	LBMA PM fix (spot)	—

Constructed Treasury and REIT series. Three legs have no investable fund reaching back to 1980, so their pre-fund segment is constructed from a published index and then spliced (tax-free, as above) into the real fund.

- **Long Treasuries** (to May 1986) and **cash / short Treasuries** (to Oct 1991) are built from the published constant-maturity Treasury yield — the 30-year for long, the 13-week T-bill for cash. Each day we hold a par Treasury at that maturity and reprice it at the next day's yield; the coupon accrual is the taxable income and the price change is capital. The matching fund's expense ratio (0.2%/yr) is applied as a daily drag over the constructed segment, the same way the gold fee is applied. After the splice the real fund's net-of-fee returns apply.
- **REITs** (to Nov 1986) use the FTSE Nareit All-Equity monthly total-return index;[10] each month's total return is distributed across that month's trading days, with the income component split out as taxable, net of a REIT-fund expense ratio (0.7%/yr), then spliced into the FRESX→VGSIX fund chain. *Limitation:* no daily REIT series exists before 1986, so the 1980–86 daily path is interpolated from monthly data and does not carry day-to-day volatility. This is the one leg whose pre-fund daily path is synthetic at the daily frequency.

Pre-1996 international and technology use proxy *funds* of the same class (no construction). The strategies that depend on these earlier-than-fund series carry the corresponding caveat.

Income and dividends. Interest and dividends come from each fund's actual per-share distribution records, captured at their real dates and amounts; for the constructed Treasury and REIT segments, the income component of the construction above. These drive the income tax below.

Costs. The gold leg carries a **0.4% annual management fee** (a physical-bullion ETF expense ratio), accrued daily as a drag on its value; it is an expense, not a sale, so it realizes no capital gain. The constructed long-Treasury, cash, and REIT segments carry the expense ratio of the fund they splice into (0.2% for the Treasuries, 0.7% for REITs), applied the same way over the pre-fund segment. Every purchase and every sale pays a **0.1% transaction cost** — each monthly contribution, each hard rebalance, each momentum switch, and each sale made to raise a tax payment. The no-rebalance and soft mixes trade only incoming contributions and bear this lightly; the hard-rebalanced and momentum strategies turn over more and bear it most. All cost rates are configurable parameters.

[10] FTSE Nareit U.S. Real Estate Index Series, monthly historical returns, Nareit (reit.com).

Contributions and window. The investor contributes 20% of gross salary on the first trading day on or after the 15th of each month, with salary set by the 75$^{\text{th}}$-percentile anchors of §1. All strategies run on **1980–2026**.

Monte Carlo. Alternative histories are built by a block bootstrap: contiguous blocks of random length (22 to 600 trading days) are drawn with replacement from the real history and stitched together until the window is filled. Each day's price return and its dividend yield are resampled *together*, so the income structure travels with the price structure. We draw 1,000 paths from a fixed seed, and re-ran the full Monte Carlo under a second independent seed; the efficient set (Section 5) is unchanged in both accounts, with only strategies sitting at the −30% threshold moving across it.

Dividends in the resampled paths. Real dividends are discrete events — monthly for bonds and Treasuries, quarterly for stocks and the REIT, semiannual for international, none for gold. Shuffling days makes the *number* of ex-dates in a synthetic year random: a year can carry more dividend events than the asset ever pays, or fewer. To keep the paths realistic, we changed the dividend event days to match each asset's real payment cadence. For each asset-year we sum the dividend the strategy actually collected — yield times shares held times price, day by day across the year — and re-lay that received total as the asset's usual number of evenly spaced, equal-dollar payments, backing out the per-day yield from the share balance and price where each lands. Because receipt depends on the share balance, which the contributions grow through the year, the re-bin is applied in-run as two passes: the first records the share path, the second re-spaces against it. A year that received nothing stays empty: a sampled year can genuinely collect no dividends — QQQ paid none until 2003, gold none ever — so we leave it at zero rather than manufacture one. Because we did not change the annual dividend amounts in these paths, our changes do not impact the actual returns or taxes in either account.

Taxes — and why they are directional. The tax model below applies to a fully taxable account. The report also runs every strategy in a fully tax-free account (Roth or 401k), where these taxes are waived; the account-aware comparison covers the two. *Income* is taxed each year: bond and Treasury interest at 20% (Treasury interest is largely state-exempt, which lowers its effective rate), stock dividends and REIT distributions at 25%, gold nothing. It is accrued per asset by calendar year and paid the following January by selling that same asset. *Realized capital gains* — from the hard-reset rebalancing (a variant of every static mix here) and the momentum strategy's switches — are taxed at 25% the same way; the soft-rebalanced and buy-and-hold strategies never sell to rebalance, so their only tax is on income.

Real taxation is far more intricate than this. Brackets rise with income, states differ, long- and short-term gains run on separate schedules, dividends split into qualified and non-qualified, the net investment income tax applies above certain thresholds, and holding an asset in a taxable versus a tax-deferred account changes everything. We deliberately use flat rates and two stylized account types — fully taxable and fully tax-free. The after-tax figures should therefore be read as **directional**: they show the *ordering* of the strategies and the rough *magnitude* of the tax drag — which mixes are hit hardest — not a precise dollar outcome for any individual.

B Calendar-Year Returns and Diversification, 1980–2026

Annual total returns for each leg (stocks and bonds include dividends; gold is price) and for the 30/30/40 mix. The blend column is the soft 30/30/40's **time-weighted return (TWR)** each year: from the strategy's actual value path, we remove each day's contribution before measuring the daily return, then compound those over the year. (1980 is measured from January 15 and 2026 is year-to-date through June 1.) The blend is a weighted combination of the legs, so its return in any year is mechanically the sum of the legs' returns at their weights – which is what lets the other legs cushion a fall in one.

Year	Stocks	Gold	Bonds	30/30/40
1980	24.4	−13.8	6.8	7.9
1981	−8.6	−32.6	10.8	−9.7
1982	19.3	14.9	26.5	20.9
1983	17.1	−16.3	3.2	0.0
1984	3.7	−19.4	11.4	−1.7
1985	22.6	6.0	17.9	14.4
1986	9.3	19.0	14.1	13.1
1987	4.7	24.5	1.6	9.0
1988	16.3	−15.3	7.4	1.4
1989	31.4	−2.8	13.7	12.8
1990	−3.3	−3.1	8.7	0.4
1991	30.2	−8.6	15.3	11.4
1992	8.2	−5.7	7.3	2.7
1993	9.1	17.7	9.7	10.9
1994	1.2	−2.2	−2.6	−2.2
1995	37.4	1.0	18.3	17.5
1996	22.9	−4.6	3.6	6.6
1997	33.2	−21.4	9.5	9.1
1998	28.6	−0.8	8.6	14.0
1999	21.1	0.9	−0.7	8.4
2000	−9.1	−5.4	11.4	−1.9
2001	−12.0	0.7	8.4	−2.4
2002	−22.2	25.6	8.4	0.1

Year	Stocks	Gold	Bonds	30/30/40
2003	28.5	19.9	3.9	14.8
2004	10.7	4.6	4.3	5.8
2005	4.8	17.8	2.4	7.1
2006	15.6	23.2	4.3	13.2
2007	5.4	31.9	7.0	14.3
2008	−37.0	4.3	5.1	−7.5
2009	26.5	25.0	6.0	17.2
2010	14.9	29.2	6.3	17.5
2011	2.0	8.9	7.6	6.5
2012	15.8	8.3	4.1	7.9
2013	32.2	−27.3	−2.2	−6.7
2014	13.5	0.1	5.8	5.7
2015	1.2	−12.1	0.3	−3.9
2016	11.8	8.1	2.5	6.8
2017	21.7	12.7	3.5	12.0
2018	−4.5	−0.9	−0.1	−2.4
2019	31.3	18.4	8.6	19.1
2020	18.2	24.6	7.6	16.1
2021	28.5	−4.3	−1.7	8.7
2022	−18.2	0.4	−13.2	−12.0
2023	26.1	14.6	5.6	16.4
2024	24.8	25.5	1.1	18.5
2025	17.7	67.4	6.7	30.4
2026	11.5	1.9	0.4	5.2

The diversification benefit, in the years it mattered. 60/40 had eight down years over the window. The table below sets them beside gold's return that year and the diversified 30/30/40:

Year	Gold	60/40	30/30/40
1981	−32.6	−1.9	−9.7
1994	−2.2	−1.3	−2.2
2000	−5.4	−3.6	−1.9
2001	0.7	−5.4	−2.4
2002	25.6	−10.7	0.1
2008	4.3	−21.3	−7.5
2018	−0.9	−3.6	−2.4
2022	0.4	−17.4	−12.0

In the three worst years for a stock/bond portfolio – 2008 (−21%), 2002 (−11%), and 2022 (−17%) – gold was flat or up, and the diversified 30/30/40 fell much less (−7.5%, +0.1%, −12.0%). 2022 is the pointed case: it is the one year in the window stocks *and* bonds fell together, the

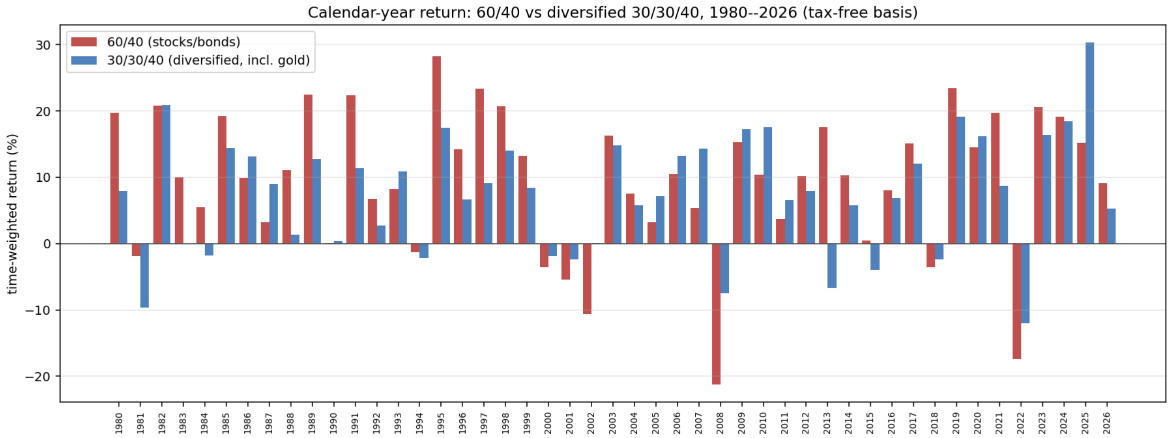

Figure 3: Calendar-year return, 60/40 (stocks/bonds) versus diversified 30/30/40 (with gold), 1980–2026. In the years the stock/bond mix fell hardest, the diversified mix fell less.

scenario a 60/40 has no defense against, and the gold holding is what cushioned the diversified mix. The benefit is not automatic every year: in 1981 gold fell 33% and the diversified mix fell more than 60/40; and the diversifier matters – All Weather's long Treasuries cushioned 2008 (it fell only −1.8%) but deepened 2022 (it fell −19.1%, worse than 60/40), because long-duration bonds rallied in the 2008 flight to quality and crashed in the 2022 rate spike. The 30/30/40 mix shown here is *not* on the efficient frontier – it is dominated on the long-run wealth-versus-pain trade – but it illustrates the point plainly: spreading risk across assets that do not all fall together cushions the years that hurt the most. No one can predict the future; but a repeat of a 2022-type year would, on this evidence, be expected to hurt a concentrated stock/bond mix more than a gold-holding diversified one.

Worst drawdown by asset class. The single deepest peak-to-trough fall each asset suffered over the window (daily total return) is uneven, and the unevenness matters for what follows:

Asset class	Worst drawdown	Period	Underwater
US stocks	−55%	2007–2009	5 yr
Intermediate bonds	−19%	2020–2022	6 yr*
Long Treasuries	−46%	2020–2023	6 yr*
Gold	−70%	1980–1999	28 yr
REITs	−73%	2007–2009	6 yr

*Still below the prior peak at the window's end (June 2026). US stocks, REITs, gold, and long Treasuries each suffered a *severe* drawdown (−46% to −73%) somewhere in the window. **Intermediate bonds did not** — their worst was −19%, in the 2022 rate spike. The 1980–2026 record is, for the intermediate-bond leg, essentially one long bull market as rates fell from double digits toward zero. Section 7 takes up what that means for the strategies that lean on it.

C Income-Path Robustness

The base case assumes a rising 75$^{\text{th}}$-percentile salary. To check whether the income path changes the conclusions, we re-run a representative set of strategies under four trajectories that hold the *same* lifetime contribution (about $532k) but differ in *when* it arrives: *base* (the rising-salary path), *flat* (equal every month), *front-loaded* (more early), and *late-bloomer* (more late). Figures are the realized 1980–2026 backtest, taxable account, final wealth in $M.

Strategy	Base	Flat	Front-loaded	Late-bloomer
30/30/40 soft	3.16	4.79	7.09	2.58
All Weather soft	3.01	4.81	7.40	2.32
Permanent Portfolio (hard)	2.17	3.07	4.33	1.81
60/40 soft	4.10	6.84	10.92	2.99
60/40 hard	3.24	5.09	7.70	2.47
80/20 soft	5.35	9.12	14.56	3.79
Aggressive soft	6.16	9.45	14.89	5.07

The level of wealth depends on the timing: contributions made earlier spend longer compounding, so the front-loaded path ends about 2.5–3× richer than the late-bloomer path for the same total contributed. The ordering of strategies is broadly stable – the equity-heavy mixes lead and the diversified mixes trail under every trajectory – with reshuffling only among strategies whose wealth is within a few percent of each other (for example 30/30/40 and All Weather).

D Missed Contributions

The base case contributes every month. To check sensitivity to missed contributions — a job loss, an emergency, a month the money went elsewhere — we skip **25% of monthly contributions at random**; we run 500 independent skip patterns. Figures are the realized 1980–2026 backtest, taxable account.

	Every month		Skips 25%		
Strategy	Final	Max DD	Mean final	Ratio [10–90th]	Max DD
30/30/40 soft	$3.16M	−21%	$2.37M	0.75 [0.72, 0.77]	−21%
All Weather soft	$3.01M	−23%	$2.26M	0.75 [0.72, 0.78]	−23%
Permanent Portfolio (hard)	$2.17M	−19%	$1.63M	0.75 [0.73, 0.77]	−19%
60/40 soft	$4.10M	−32%	$3.08M	0.75 [0.72, 0.78]	−32%
60/40 hard	$3.24M	−33%	$2.44M	0.75 [0.72, 0.78]	−33%
80/20 soft	$5.35M	−43%	$4.01M	0.75 [0.72, 0.78]	−43%
Aggressive soft	$6.16M	−63%	$4.61M	0.75 [0.72, 0.77]	−63%

Skipping a quarter of the contributions reduces final wealth by about a quarter (the ratio is near 0.75 for every strategy, with a 10–90th-percentile band of roughly 0.72–0.78), and leaves the maximum drawdown essentially unchanged. The proportional effect is the same across strategies, so the ranking is unchanged; the cost of the missed contributions is a smaller balance, not a different conclusion.

E Horizon Robustness

The base case accumulates for **46 years** (age 21 to 67). Not every investor saves that long, so we test whether the efficient set holds at shorter horizons — **31, 35, 38, 40, 44, and 46 years**. The Monte Carlo paths are drawn from the same full 1980–2026 return pool as everywhere else. A *static* mix holds fixed weights, so the first H years of a 46-year path is itself a valid H-year run, and we slice it. A *glide* is horizon-specific by construction — a 46-year 90/10→30/70 glide is only two-thirds of the way to its endpoint at year 31 — so each glide is rebuilt to *complete at the horizon* (the continuous glides reach their endpoint at H; the stepped glides place their transitions at the same fractions of the shorter span) and run separately at each horizon. Terminal pain is measured over the final five years before each cutoff. At each horizon and account we recompute the within-cap ($\leq -30\%$ terminal pain) efficient frontier; the table marks which strategies stay efficient.

Within-cap efficient strategies by accumulation length (years) and account (1980 return pool, 1,000 histories)

	Taxable						Tax-free					
Strategy	31	35	38	40	44	46	31	35	38	40	44	46
TDF 90/10→30/70 hard	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
TDF 90/10→30/30/40 soft	✓											
Permanent Portfolio (hard)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Permanent Portfolio (soft)	✓											
glide A (hard)	✓	✓					✓					
All Weather (no rebal)	✓	✓	✓	✓	✓	✓				✓		✓
50/25/25 (soft)	✓	✓	✓	✓								
TDF 90/10→30/70 soft	✓	✓	✓				✓	✓	✓	✓		
All Weather (soft)	✓	✓	✓	✓	✓	✓						
TDF 90/10→30/30/40 hard		✓	✓	✓	✓							
40/25/35 (soft)			✓	✓	✓	✓						
glide D (soft)				✓	✓	✓						
60/40 (hard)								✓	✓	✓	✓	✓
30/30/40 (hard)										✓		

Two strategies — the **90/10→30/70 target-date glide (hard)** and the **Permanent Portfolio (hard)** — are efficient at every horizon in both accounts; no other strategy is. The rest of the set changes with the horizon. The glide's terminal pain is lower at the shorter horizons, not higher: an investor retiring sooner reaches its 70%-bond endpoint earlier, so its terminal pain falls to about −18%. This is the bond effect of Section 7 — the glide's low terminal pain comes from a large intermediate-bond weight, and the record never tested that weight against a serious bond fall.

The diversified mixes by horizon. Median wealth and terminal pain for the stocks/gold/bonds family (taxable account; median wealth $M / terminal pain %), each in its three rebalancing variants:

Strategy	31yr	35yr	38yr	40yr	44yr	46yr
30/30/40 soft	0.8/−21	1.1/−22	1.5/−23	1.8/−23	2.5/−26	3.0/−25
30/30/40 hard	0.7/−20	1.0/−20	1.2/−20	1.4/−20	1.9/−20	2.2/−20
30/30/40 no-reb	0.8/−25	1.2/−27	1.7/−27	2.0/−28	2.9/−30	3.5/−30
40/30/30 soft	0.9/−25	1.3/−26	1.7/−26	2.0/−27	2.9/−29	3.5/−29
40/30/30 hard	0.8/−23	1.1/−23	1.4/−23	1.6/−23	2.2/−24	2.6/−24
40/30/30 no-reb	0.9/−29	1.4/−31	1.9/−31	2.3/−31	3.4/−33	4.1/−34
40/25/35 soft	0.9/−24	1.2/−25	1.7/−25	2.0/−26	2.9/−28	3.5/−28
40/25/35 hard	0.8/−21	1.1/−22	1.4/−21	1.6/−21	2.2/−22	2.6/−23
40/25/35 no-reb	0.9/−29	1.4/−31	1.9/−31	2.3/−31	3.4/−33	4.0/−34
50/25/25 soft	0.9/−27	1.4/−28	1.9/−28	2.3/−29	3.3/−31	4.0/−31
50/25/25 hard	0.8/−26	1.2/−26	1.6/−26	1.9/−25	2.6/−26	3.1/−27
50/25/25 no-reb	1.0/−33	1.5/−35	2.1/−34	2.5/−34	3.8/−36	4.6/−37

Hard rebalancing holds terminal pain within the −30% cap across all four mixes at every horizon, and roughly flat across horizons. The soft versions stay within the cap as well, except 50/25/25, which reaches −31% by year 44. The never-rebalance versions breach at the longer horizons: with no trimming, the equity weight drifts up over time, so terminal pain deepens with both equity and horizon — 40/30/30, 40/25/35, and 50/25/25 reach −34% to −37% by year 46, while 30/30/40 sits at −30%. The tax-free account is similar, with higher wealth and comparable terminal pain. None of these mixes matches the bond-heavy glide on wealth per unit of terminal pain, but each splits its defensive weight between gold and bonds rather than bonds alone.

Does a pick stay viable if you work longer than planned? Suppose you choose a fixed-weight mix — one held to constant target weights, not a glide — that sits within the −30% cap for a retirement planned at year H, then keep saving to year 46. Does its terminal pain cross the cap? Almost none do. The one exception is 50/25/25 (soft), and it crosses only marginally: −29% at the shorter horizons, −31% at year 46. Every mix well inside the cap stays inside; the Permanent Portfolio (hard), for one, holds near −17% throughout. A glide is a separate case: it is built for a chosen horizon, so working longer means rebuilding it to a later endpoint, not carrying the old one on.

One caveat covers the whole appendix. Every horizon here is the same 1980–2026 return history read at a different stopping point, so these results show the conclusions hold whether you accumulate for 31 years or 46 — not that they would hold in a future unlike 1980–2026. No resampling of a single history can establish that (Section 9).

F All Strategies, Both Account Types

All 56 strategies run in full, across both account types (fully taxable and fully tax-free). Each strategy is shown under four outcomes: the actual realized backtest, and three illustrative Monte Carlo paths (the median by return, the 10th percentile by return, and the 10th percentile by maximum drawdown).

Each column is one whole path. The eight figures stacked in a cell — wealth, IRR, full maximum drawdown, final-five-year drawdown, peak-to-trough years, recovery, Sharpe, and Calmar — all come from that single outcome; nothing is stitched together across paths.

Two terminal-pain figures appear, and they are not the same thing. The fourth line in each cell is *that path's own* drawdown in its final five years. The figure printed under each strategy *name* is the strategy's *terminal pain*, the 10th-percentile final-five-year drawdown across all 1,000 paths — a percentile of the whole distribution, so it lives in no single path.

1980, taxable. The figure under each strategy name is its *terminal pain*, the 10th-percentile final-5yr drawdown across all 1,000 paths. Each strategy spans eight rows, one per metric. *Final-5yr DD* is *that path's own* terminal drawdown; *Recovery* is years to recover, or residual DD at the window end.

Appendix G — 1980, taxable

Strategy	Metric	Actual	MC median by return	MC 10th pct by return	MC 10th pct by max DD
	Wealth	3.16M	3.10M	1.82M	3.30M
	IRR	7.8%	7.7%	5.7%	8.0%
	Max DD (full)	−21.1%	−18.0%	−27.1%	−28.8%
30/30/40 soft term. pain −25%	Final-5yr DD	−17.3%	−18.0%	−27.1%	−28.8%
	Peak-to-trough	0.7y	0.8y	0.8y	0.4y
	Recovery	1.5y	4.2y	1.7y	−20% end
	Sharpe	0.47	0.49	0.30	0.52
	Calmar	0.37	0.43	0.21	0.28

Appendix G — 1980, taxable

Strategy	Metric	Actual	MC median by return	MC 10th pct by return	MC 10th pct by max DD
30/30/40 no-reb term. pain −30%	Wealth	3.57M	3.63M	1.93M	2.24M
	IRR	8.3%	8.3%	6.0%	6.5%
	Max DD (full)	−24.3%	−27.1%	−25.6%	−35.3%
	Final-5yr DD	−19.4%	−17.6%	−14.2%	−29.0%
	Peak-to-trough	0.5y	1.7y	0.5y	1.3y
	Recovery	1.4y	5.6y	1.7y	3.6y
	Sharpe	0.48	0.52	0.31	0.36
	Calmar	0.34	0.31	0.23	0.19
30/30/40 hard term. pain −21%	Wealth	2.56M	2.22M	1.51M	2.30M
	IRR	7.0%	6.5%	5.0%	6.6%
	Max DD (full)	−20.2%	−19.9%	−24.3%	−22.7%
	Final-5yr DD	−17.5%	−5.4%	−12.7%	−18.2%
	Peak-to-trough	0.7y	0.6y	2.5y	1.9y
	Recovery	1.5y	3.7y	3.4y	3.8y
	Sharpe	0.43	0.46	0.33	0.44
	Calmar	0.35	0.33	0.21	0.29
40/25/35 soft term. pain −28%	Wealth	3.60M	3.57M	2.00M	2.92M
	IRR	8.3%	8.3%	6.1%	7.5%
	Max DD (full)	−23.4%	−20.8%	−27.9%	−32.3%
	Final-5yr DD	−18.6%	−11.3%	−11.0%	−32.3%
	Peak-to-trough	0.7y	0.5y	3.1y	1.1y
	Recovery	1.5y	2.0y	4.8y	−23% end
	Sharpe	0.50	0.42	0.46	0.52
	Calmar	0.35	0.40	0.22	0.23
40/25/35 no-reb term. pain −34%	Wealth	4.08M	4.22M	2.07M	2.45M
	IRR	8.7%	8.8%	6.2%	6.9%
	Max DD (full)	−28.4%	−28.7%	−32.8%	−39.6%
	Final-5yr DD	−20.7%	−28.7%	−32.8%	−35.7%
	Peak-to-trough	0.5y	0.9y	1.4y	1.7y
	Recovery	1.5y	−29% end	2.3y	4.1y
	Sharpe	0.48	0.49	0.37	0.40
	Calmar	0.31	0.31	0.19	0.17
40/25/35 hard term. pain −23%	Wealth	2.92M	2.64M	1.71M	1.40M
	IRR	7.5%	7.2%	5.5%	4.7%
	Max DD (full)	−24.1%	−17.2%	−22.8%	−26.0%
	Final-5yr DD	−18.9%	−9.1%	−13.4%	−26.0%
	Peak-to-trough	0.5y	1.4y	0.7y	0.5y
	Recovery	1.4y	2.2y	1.5y	−3% end
	Sharpe	0.47	0.54	0.33	0.23
	Calmar	0.31	0.42	0.24	0.18
50/25/25 soft term. pain −31%	Wealth	4.21M	4.14M	2.13M	3.26M
	IRR	8.8%	8.8%	6.4%	7.9%
	Max DD (full)	−27.0%	−42.8%	−32.2%	−36.2%
	Final-5yr DD	−19.5%	−42.8%	−27.0%	−25.4%
	Peak-to-trough	0.7y	1.1y	1.1y	2.1y
	Recovery	1.6y	−18% end	5.7y	4.5y
	Sharpe	0.48	0.47	0.25	0.31
	Calmar	0.33	0.21	0.20	0.22
50/25/25 no-reb term. pain −37%	Wealth	4.70M	4.88M	2.23M	3.69M
	IRR	9.2%	9.3%	6.5%	8.4%
	Max DD (full)	−33.4%	−39.3%	−34.1%	−43.5%
	Final-5yr DD	−21.5%	−39.3%	−26.3%	−22.1%
	Peak-to-trough	1.1y	0.5y	0.5y	1.2y
	Recovery	2.4y	1.4y	3.1y	2.5y
	Sharpe	0.46	0.45	0.30	0.35
	Calmar	0.28	0.24	0.19	0.19

Appendix G — 1980, taxable

Strategy	Metric	Actual	MC median by return	MC 10th pct by return	MC 10th pct by max DD
50/25/25 hard term. pain −27%	Wealth	3.53M	3.12M	1.83M	3.49M
	IRR	8.2%	7.8%	5.7%	8.2%
	Max DD (full)	−28.8%	−40.1%	−22.5%	−31.0%
	Final-5yr DD	−19.8%	−38.9%	−12.8%	−19.7%
	Peak-to-trough	0.5y	1.1y	1.3y	0.8y
	Recovery	1.5y	3.1y	2.8y	1.8y
	Sharpe	0.46	0.48	0.36	0.43
	Calmar	0.29	0.19	0.26	0.26
40/30/30 soft term. pain −29%	Wealth	3.73M	3.63M	1.97M	3.68M
	IRR	8.4%	8.3%	6.0%	8.4%
	Max DD (full)	−24.6%	−23.1%	−29.0%	−33.0%
	Final-5yr DD	−18.3%	−6.2%	−15.2%	−26.1%
	Peak-to-trough	0.7y	0.5y	1.2y	0.5y
	Recovery	1.6y	1.6y	3.7y	2.1y
	Sharpe	0.46	0.61	0.40	0.45
	Calmar	0.34	0.36	0.21	0.25
40/30/30 no-reb term. pain −34%	Wealth	4.19M	4.25M	2.05M	5.50M
	IRR	8.8%	8.9%	6.2%	9.8%
	Max DD (full)	−29.1%	−41.1%	−36.2%	−39.9%
	Final-5yr DD	−20.5%	−38.0%	−27.5%	−11.2%
	Peak-to-trough	0.5y	1.4y	1.6y	4.9y
	Recovery	1.5y	4.2y	8.1y	7.3y
	Sharpe	0.46	0.45	0.23	0.55
	Calmar	0.30	0.22	0.17	0.24
40/30/30 hard term. pain −24%	Wealth	3.08M	2.62M	1.65M	2.34M
	IRR	7.7%	7.1%	5.4%	6.7%
	Max DD (full)	−25.0%	−27.5%	−18.7%	−27.1%
	Final-5yr DD	−18.5%	−27.5%	−15.1%	−27.1%
	Peak-to-trough	0.5y	0.6y	1.0y	0.8y
	Recovery	1.4y	1.6y	5.7y	−5% end
	Sharpe	0.44	0.33	0.34	0.42
	Calmar	0.31	0.26	0.29	0.25
60/40 soft term. pain −34%	Wealth	4.10M	4.53M	2.29M	1.85M
	IRR	8.7%	9.1%	6.6%	5.8%
	Max DD (full)	−31.6%	−41.3%	−31.5%	−39.1%
	Final-5yr DD	−22.2%	−21.8%	−10.1%	−38.1%
	Peak-to-trough	1.4y	0.8y	0.9y	1.6y
	Recovery	3.0y	3.6y	1.8y	3.7y
	Sharpe	0.54	0.49	0.37	0.28
	Calmar	0.28	0.22	0.21	0.15
60/40 no-reb term. pain −40%	Wealth	4.78M	5.48M	2.36M	3.26M
	IRR	9.3%	9.7%	6.7%	7.9%
	Max DD (full)	−39.1%	−27.4%	−38.5%	−47.4%
	Final-5yr DD	−23.0%	−15.8%	−26.4%	−18.2%
	Peak-to-trough	1.4y	0.3y	1.0y	3.1y
	Recovery	3.2y	0.7y	2.1y	4.8y
	Sharpe	0.49	0.49	0.22	0.32
	Calmar	0.24	0.36	0.17	0.17
60/40 hard term. pain −29%	Wealth	3.24M	3.74M	2.10M	1.86M
	IRR	7.9%	8.4%	6.3%	5.8%
	Max DD (full)	−32.7%	−21.5%	−30.7%	−34.8%
	Final-5yr DD	−22.5%	−10.8%	−18.3%	−34.8%
	Peak-to-trough	1.4y	0.2y	1.0y	1.4y
	Recovery	2.5y	1.3y	2.0y	−21% end
	Sharpe	0.50	0.61	0.20	0.23
	Calmar	0.24	0.39	0.20	0.17

Appendix G — 1980, taxable

Strategy	Metric	Actual	MC median by return	MC 10th pct by return	MC 10th pct by max DD
PermPort soft term. pain −23%	Wealth	2.77M	2.87M	1.83M	5.81M
	IRR	7.3%	7.5%	5.8%	9.9%
	Max DD (full)	−18.7%	−10.9%	−17.3%	−25.2%
	Final-5yr DD	−18.7%	−7.9%	−17.0%	−13.8%
	Peak-to-trough	0.8y	1.1y	1.6y	0.7y
	Recovery	2.0y	1.9y	4.1y	1.5y
	Sharpe	0.52	0.59	0.35	0.67
	Calmar	0.39	0.68	0.33	0.39
PermPort no-reb term. pain −27%	Wealth	3.13M	3.43M	1.93M	1.32M
	IRR	7.8%	8.1%	6.0%	4.4%
	Max DD (full)	−21.2%	−38.5%	−27.1%	−31.0%
	Final-5yr DD	−21.2%	−38.5%	−17.7%	−21.0%
	Peak-to-trough	0.8y	1.1y	1.2y	1.3y
	Recovery	2.1y	−16% end	4.7y	3.1y
	Sharpe	0.53	0.47	0.38	0.19
	Calmar	0.37	0.21	0.22	0.14
PermPort hard term. pain −18%	Wealth	2.17M	2.10M	1.48M	1.82M
	IRR	6.4%	6.3%	4.9%	5.7%
	Max DD (full)	−19.3%	−23.3%	−17.8%	−19.7%
	Final-5yr DD	−19.3%	−10.9%	−17.8%	−5.5%
	Peak-to-trough	0.9y	1.2y	0.9y	1.2y
	Recovery	2.4y	1.7y	1.9y	1.6y
	Sharpe	0.45	0.39	0.49	0.34
	Calmar	0.33	0.27	0.28	0.29
Conservative soft term. pain −30%	Wealth	2.79M	2.93M	1.68M	3.10M
	IRR	7.4%	7.5%	5.4%	7.7%
	Max DD (full)	−31.0%	−24.6%	−24.0%	−37.0%
	Final-5yr DD	−20.0%	−23.9%	−19.6%	−26.3%
	Peak-to-trough	0.5y	1.1y	2.0y	0.5y
	Recovery	1.5y	1.5y	4.4y	5.5y
	Sharpe	0.42	0.49	0.36	0.41
	Calmar	0.24	0.31	0.23	0.21
Conservative no-reb term. pain −34%	Wealth	2.97M	3.43M	1.76M	8.78M
	IRR	7.6%	8.1%	5.6%	11.3%
	Max DD (full)	−36.3%	−24.8%	−33.0%	−43.0%
	Final-5yr DD	−22.3%	−11.8%	−15.7%	−35.1%
	Peak-to-trough	0.5y	0.6y	0.5y	0.8y
	Recovery	1.9y	1.9y	1.5y	2.9y
	Sharpe	0.41	0.31	0.37	0.65
	Calmar	0.21	0.33	0.17	0.26
Conservative hard term. pain −26%	Wealth	2.40M	2.18M	1.38M	1.12M
	IRR	6.8%	6.4%	4.6%	3.7%
	Max DD (full)	−30.7%	−29.8%	−20.2%	−31.7%
	Final-5yr DD	−19.7%	−29.8%	−10.1%	−7.8%
	Peak-to-trough	0.5y	0.7y	1.0y	0.9y
	Recovery	1.4y	−18% end	2.3y	4.4y
	Sharpe	0.39	0.48	0.28	0.09
	Calmar	0.22	0.22	0.23	0.12
Aggressive soft term. pain −55%	Wealth	6.16M	6.25M	1.91M	4.33M
	IRR	10.1%	10.2%	5.9%	8.9%
	Max DD (full)	−63.5%	−55.2%	−42.4%	−62.8%
	Final-5yr DD	−28.5%	−23.9%	−42.4%	−12.5%
	Peak-to-trough	2.5y	2.7y	1.0y	1.5y
	Recovery	6.9y	5.8y	3.7y	10.9y
	Sharpe	0.33	0.35	0.24	0.37
	Calmar	0.16	0.18	0.14	0.14

Appendix G — 1980, taxable

Strategy	Metric	Actual	MC median by return	MC 10th pct by return	MC 10th pct by max DD
Aggressive no-reb term. pain −59%	Wealth	8.04M	7.72M	2.09M	8.85M
	IRR	11.0%	10.9%	6.3%	11.3%
	Max DD (full)	−72.3%	−47.2%	−58.7%	−67.3%
	Final-5yr DD	−31.3%	−21.4%	−41.8%	−33.3%
	Peak-to-trough	2.5y	3.4y	3.1y	1.9y
	Recovery	10.5y	6.7y	8.8y	13.8y
	Sharpe	0.35	0.41	0.21	0.43
	Calmar	0.15	0.23	0.11	0.17
Aggressive hard term. pain −42%	Wealth	4.73M	3.20M	1.43M	2.52M
	IRR	9.2%	7.9%	4.8%	7.0%
	Max DD (full)	−41.9%	−40.2%	−44.3%	−49.0%
	Final-5yr DD	−25.1%	−31.9%	−20.6%	−23.8%
	Peak-to-trough	2.6y	1.5y	3.1y	3.3y
	Recovery	4.7y	2.5y	7.7y	5.0y
	Sharpe	0.34	0.30	0.15	0.27
	Calmar	0.22	0.20	0.11	0.14
US/Intl 50/50 soft term. pain −51%	Wealth	3.98M	4.44M	1.65M	1.30M
	IRR	8.6%	9.0%	5.4%	4.4%
	Max DD (full)	−57.9%	−42.9%	−53.7%	−61.0%
	Final-5yr DD	−25.5%	−11.4%	−52.6%	−48.7%
	Peak-to-trough	1.4y	1.6y	0.8y	7.1y
	Recovery	5.2y	3.7y	2.4y	17.8y
	Sharpe	0.34	0.43	0.12	0.18
	Calmar	0.15	0.21	0.10	0.07
US/Intl 50/50 no-reb term. pain −50%	Wealth	4.55M	5.15M	1.94M	2.08M
	IRR	9.1%	9.5%	6.0%	6.3%
	Max DD (full)	−56.8%	−54.0%	−48.4%	−60.1%
	Final-5yr DD	−25.1%	−20.0%	−33.6%	−50.9%
	Peak-to-trough	1.4y	1.0y	2.4y	4.5y
	Recovery	4.9y	3.9y	10.3y	−26% end
	Sharpe	0.35	0.38	0.26	0.26
	Calmar	0.16	0.18	0.12	0.10
US/Intl 50/50 hard term. pain −53%	Wealth	3.32M	3.60M	1.43M	3.40M
	IRR	8.0%	8.3%	4.8%	8.1%
	Max DD (full)	−57.7%	−45.6%	−48.9%	−61.8%
	Final-5yr DD	−25.9%	−33.7%	−29.5%	−33.6%
	Peak-to-trough	1.4y	11.0y	1.4y	8.7y
	Recovery	5.2y	11.8y	2.9y	11.4y
	Sharpe	0.30	0.29	0.05	0.25
	Calmar	0.14	0.18	0.10	0.13
All Weather soft term. pain −25%	Wealth	3.01M	3.46M	2.09M	2.91M
	IRR	7.6%	8.1%	6.3%	7.5%
	Max DD (full)	−23.4%	−24.5%	−24.9%	−27.7%
	Final-5yr DD	−23.4%	−22.2%	−24.9%	−25.4%
	Peak-to-trough	0.8y	1.8y	0.8y	1.0y
	Recovery	2.4y	4.8y	2.7y	2.5y
	Sharpe	0.58	0.56	0.43	0.57
	Calmar	0.33	0.33	0.25	0.27
All Weather no-reb term. pain −28%	Wealth	3.30M	3.97M	2.14M	3.61M
	IRR	8.0%	8.6%	6.4%	8.3%
	Max DD (full)	−24.2%	−22.8%	−18.1%	−33.1%
	Final-5yr DD	−24.2%	−12.5%	−9.0%	−24.1%
	Peak-to-trough	0.8y	0.2y	2.0y	1.0y
	Recovery	2.2y	1.6y	3.2y	3.6y
	Sharpe	0.57	0.52	0.45	0.51
	Calmar	0.33	0.38	0.35	0.25

Appendix G — 1980, taxable

Strategy	Metric	Actual	MC median by return	MC 10th pct by return	MC 10th pct by max DD
All Weather hard term. pain −21%	Wealth	2.35M	2.60M	1.72M	2.15M
	IRR	6.7%	7.1%	5.5%	6.4%
	Max DD (full)	−25.8%	−16.3%	−28.9%	−25.2%
	Final-5yr DD	−25.8%	−7.4%	−21.7%	−10.9%
	Peak-to-trough	0.8y	1.0y	4.8y	1.1y
	Recovery	2.7y	1.9y	6.3y	4.0y
	Sharpe	0.51	0.52	0.46	0.56
	Calmar	0.26	0.44	0.19	0.25
Three-fund soft term. pain −33%	Wealth	3.17M	3.61M	1.87M	2.88M
	IRR	7.8%	8.3%	5.8%	7.5%
	Max DD (full)	−35.7%	−27.0%	−35.0%	−39.9%
	Final-5yr DD	−22.5%	−23.2%	−18.3%	−38.1%
	Peak-to-trough	1.4y	1.0y	2.4y	2.9y
	Recovery	3.0y	3.3y	6.1y	5.9y
	Sharpe	0.49	0.46	0.27	0.35
	Calmar	0.22	0.31	0.17	0.19
Three-fund no-reb term. pain −38%	Wealth	3.70M	4.30M	1.97M	3.93M
	IRR	8.4%	8.9%	6.1%	8.6%
	Max DD (full)	−38.7%	−38.9%	−46.2%	−46.0%
	Final-5yr DD	−22.9%	−11.2%	−46.2%	−16.4%
	Peak-to-trough	1.4y	1.6y	2.1y	1.4y
	Recovery	3.2y	6.5y	−19% end	3.4y
	Sharpe	0.47	0.49	0.33	0.40
	Calmar	0.22	0.23	0.13	0.19
Three-fund hard term. pain −30%	Wealth	2.50M	2.87M	1.67M	3.18M
	IRR	7.0%	7.5%	5.4%	7.8%
	Max DD (full)	−34.3%	−35.1%	−22.4%	−35.9%
	Final-5yr DD	−22.8%	−18.2%	−15.4%	−7.9%
	Peak-to-trough	1.4y	1.1y	1.0y	3.0y
	Recovery	2.5y	3.5y	2.6y	6.1y
	Sharpe	0.45	0.46	0.33	0.46
	Calmar	0.20	0.21	0.24	0.22
70/30 soft term. pain −37%	Wealth	4.71M	5.19M	2.45M	2.88M
	IRR	9.2%	9.6%	6.9%	7.5%
	Max DD (full)	−37.2%	−32.8%	−37.0%	−42.9%
	Final-5yr DD	−22.7%	−32.8%	−19.2%	−42.9%
	Peak-to-trough	1.4y	1.4y	1.2y	1.8y
	Recovery	3.2y	3.7y	4.1y	3.6y
	Sharpe	0.51	0.47	0.36	0.47
	Calmar	0.25	0.29	0.19	0.17
70/30 no-reb term. pain −42%	Wealth	5.40M	6.09M	2.50M	10.25M
	IRR	9.7%	10.1%	7.0%	11.8%
	Max DD (full)	−43.8%	−28.5%	−45.3%	−49.7%
	Final-5yr DD	−23.5%	−13.9%	−16.0%	−41.6%
	Peak-to-trough	1.4y	2.4y	1.2y	0.8y
	Recovery	3.3y	4.3y	3.6y	3.3y
	Sharpe	0.47	0.41	0.33	0.51
	Calmar	0.22	0.35	0.15	0.24
70/30 hard term. pain −34%	Wealth	3.95M	4.47M	2.30M	1.53M
	IRR	8.6%	9.0%	6.6%	5.0%
	Max DD (full)	−38.6%	−28.7%	−35.0%	−41.0%
	Final-5yr DD	−23.1%	−14.4%	−13.2%	−24.4%
	Peak-to-trough	1.4y	1.0y	0.9y	6.9y
	Recovery	3.1y	2.8y	1.8y	10.1y
	Sharpe	0.47	0.57	0.33	0.22
	Calmar	0.22	0.32	0.19	0.12

Appendix G — 1980, taxable

Strategy	Metric	Actual	MC median by return	MC 10th pct by return	MC 10th pct by max DD
80/20 soft term. pain −40%	Wealth	5.35M	5.94M	2.58M	7.19M
	IRR	9.7%	10.0%	7.1%	10.7%
	Max DD (full)	−42.7%	−40.5%	−37.6%	−46.4%
	Final-5yr DD	−23.2%	−22.7%	−26.5%	−17.4%
	Peak-to-trough	1.4y	1.4y	1.1y	2.8y
	Recovery	3.3y	4.9y	4.1y	5.9y
	Sharpe	0.49	0.42	0.26	0.53
	Calmar	0.23	0.25	0.19	0.23
80/20 no-reb term. pain −45%	Wealth	6.02M	6.75M	2.60M	12.86M
	IRR	10.1%	10.4%	7.1%	12.6%
	Max DD (full)	−47.8%	−39.8%	−48.2%	−52.3%
	Final-5yr DD	−23.9%	−38.5%	−18.9%	−32.6%
	Peak-to-trough	1.4y	1.1y	1.2y	1.4y
	Recovery	3.5y	1.9y	4.4y	4.0y
	Sharpe	0.45	0.43	0.32	0.52
	Calmar	0.21	0.26	0.15	0.24
80/20 hard term. pain −39%	Wealth	4.83M	5.39M	2.47M	16.22M
	IRR	9.3%	9.7%	6.9%	13.3%
	Max DD (full)	−44.2%	−50.3%	−44.4%	−46.7%
	Final-5yr DD	−23.6%	−15.0%	−38.9%	−16.4%
	Peak-to-trough	1.4y	4.6y	1.5y	1.0y
	Recovery	3.3y	9.2y	2.9y	2.4y
	Sharpe	0.46	0.49	0.29	0.66
	Calmar	0.21	0.19	0.16	0.28
glA soft term. pain −47%	Wealth	4.56M	5.40M	2.11M	6.52M
	IRR	9.1%	9.7%	6.3%	10.3%
	Max DD (full)	−53.7%	−25.9%	−40.9%	−55.1%
	Final-5yr DD	−27.3%	−16.1%	−28.2%	−55.1%
	Peak-to-trough	2.5y	0.8y	2.8y	1.1y
	Recovery	7.5y	1.7y	6.0y	−46% end
	Sharpe	0.34	0.43	0.35	0.49
	Calmar	0.17	0.37	0.15	0.19
glA hard term. pain −21%	Wealth	2.65M	2.78M	1.77M	2.06M
	IRR	7.2%	7.3%	5.6%	6.2%
	Max DD (full)	−20.9%	−19.4%	−22.9%	−33.5%
	Final-5yr DD	−18.6%	−8.1%	−22.9%	−15.4%
	Peak-to-trough	0.4y	0.1y	0.9y	2.2y
	Recovery	1.8y	0.2y	−9% end	3.9y
	Sharpe	0.32	0.56	0.26	0.22
	Calmar	0.34	0.38	0.25	0.19
glB soft term. pain −48%	Wealth	5.06M	6.01M	2.19M	6.70M
	IRR	9.5%	10.1%	6.5%	10.4%
	Max DD (full)	−55.3%	−50.9%	−30.2%	−56.7%
	Final-5yr DD	−27.3%	−24.1%	−11.2%	−48.8%
	Peak-to-trough	2.5y	1.3y	1.8y	1.0y
	Recovery	7.1y	4.3y	4.2y	3.7y
	Sharpe	0.34	0.44	0.41	0.46
	Calmar	0.17	0.20	0.21	0.18
glB hard term. pain −29%	Wealth	3.57M	2.94M	1.67M	3.23M
	IRR	8.2%	7.5%	5.4%	7.9%
	Max DD (full)	−31.3%	−25.7%	−21.4%	−37.3%
	Final-5yr DD	−20.1%	−9.1%	−12.8%	−13.4%
	Peak-to-trough	0.5y	0.5y	0.1y	0.8y
	Recovery	1.5y	1.8y	0.4y	1.9y
	Sharpe	0.38	0.27	0.29	0.47
	Calmar	0.26	0.29	0.25	0.21

Appendix G — 1980, taxable

Strategy	Metric	Actual	MC median by return	MC 10th pct by return	MC 10th pct by max DD
glC soft term. pain −35%	Wealth	3.37M	3.93M	1.90M	8.51M
	IRR	8.0%	8.6%	5.9%	11.2%
	Max DD (full)	−34.6%	−47.6%	−47.1%	−48.4%
	Final-5yr DD	−21.6%	−22.3%	−17.0%	−38.3%
	Peak-to-trough	0.3y	1.1y	1.8y	1.0y
	Recovery	1.8y	4.6y	5.3y	3.2y
	Sharpe	0.44	0.32	0.38	0.63
	Calmar	0.23	0.18	0.13	0.23
glC hard term. pain −30%	Wealth	3.89M	2.97M	1.70M	4.25M
	IRR	8.6%	7.6%	5.5%	8.9%
	Max DD (full)	−34.6%	−52.3%	−30.5%	−44.6%
	Final-5yr DD	−20.2%	−16.3%	−28.4%	−11.7%
	Peak-to-trough	0.3y	1.2y	1.1y	0.5y
	Recovery	1.8y	2.7y	2.3y	1.6y
	Sharpe	0.51	0.19	0.26	0.40
	Calmar	0.25	0.15	0.18	0.20
glD soft term. pain −27%	Wealth	3.49M	3.57M	1.94M	4.99M
	IRR	8.2%	8.3%	6.0%	9.4%
	Max DD (full)	−22.8%	−22.9%	−19.8%	−32.9%
	Final-5yr DD	−18.0%	−6.4%	−18.3%	−21.6%
	Peak-to-trough	0.7y	0.5y	0.5y	2.5y
	Recovery	1.5y	1.3y	1.3y	3.3y
	Sharpe	0.48	0.50	0.39	0.51
	Calmar	0.36	0.36	0.30	0.29
glD hard term. pain −25%	Wealth	3.46M	3.09M	1.84M	2.44M
	IRR	8.1%	7.7%	5.8%	6.9%
	Max DD (full)	−21.5%	−17.3%	−29.9%	−28.1%
	Final-5yr DD	−17.1%	−6.8%	−29.9%	−20.2%
	Peak-to-trough	0.7y	0.1y	1.1y	0.6y
	Recovery	1.5y	0.3y	−29% end	2.7y
	Sharpe	0.49	0.43	0.39	0.37
	Calmar	0.38	0.45	0.19	0.24
GEM monthly term. pain −36%	Wealth	3.22M	2.36M	1.22M	1.37M
	IRR	7.9%	6.7%	4.1%	4.6%
	Max DD (full)	−33.5%	−31.3%	−33.5%	−43.7%
	Final-5yr DD	−33.4%	−25.2%	−25.9%	−30.5%
	Peak-to-trough	1.5y	0.6y	0.1y	0.4y
	Recovery	3.1y	1.9y	2.0y	2.5y
	Sharpe	0.40	0.25	0.09	0.14
	Calmar	0.24	0.22	0.12	0.10
GEM quarterly term. pain −40%	Wealth	2.55M	2.45M	1.26M	1.74M
	IRR	7.0%	6.9%	4.2%	5.6%
	Max DD (full)	−47.6%	−38.1%	−38.8%	−52.9%
	Final-5yr DD	−29.7%	−29.3%	−31.7%	−18.5%
	Peak-to-trough	0.3y	1.7y	1.9y	1.7y
	Recovery	4.1y	3.4y	3.9y	7.5y
	Sharpe	0.29	0.27	0.11	-0.01
	Calmar	0.15	0.18	0.11	0.11
TrendFollow term. pain −29%	Wealth	3.55M	3.51M	1.88M	4.46M
	IRR	8.2%	8.2%	5.9%	9.0%
	Max DD (full)	−23.8%	−28.6%	−35.6%	−33.4%
	Final-5yr DD	−18.0%	−23.8%	−35.6%	−17.3%
	Peak-to-trough	0.7y	0.5y	1.1y	1.1y
	Recovery	1.5y	1.5y	−36% end	3.8y
	Sharpe	0.44	0.40	0.34	0.54
	Calmar	0.35	0.29	0.16	0.27

Appendix G — 1980, taxable

Strategy	Metric	Actual	MC median by return	MC 10th pct by return	MC 10th pct by max DD
MeanRevert term. pain −27%	Wealth	3.45M	3.32M	1.87M	4.24M
	IRR	8.1%	8.0%	5.8%	8.9%
	Max DD (full)	−23.2%	−18.5%	−29.2%	−31.4%
	Final-5yr DD	−17.4%	−18.5%	−27.8%	−31.4%
	Peak-to-trough	0.7y	0.6y	1.2y	0.5y
	Recovery	1.5y	2.7y	2.1y	1.9y
	Sharpe	0.44	0.44	0.34	0.51
	Calmar	0.35	0.43	0.20	0.28
TF+MR term. pain −28%	Wealth	3.48M	3.41M	1.89M	3.53M
	IRR	8.2%	8.1%	5.9%	8.2%
	Max DD (full)	−23.3%	−27.1%	−26.2%	−32.4%
	Final-5yr DD	−17.6%	−20.4%	−17.4%	−27.5%
	Peak-to-trough	0.7y	1.8y	0.7y	0.5y
	Recovery	1.5y	3.5y	4.5y	1.8y
	Sharpe	0.45	0.48	0.37	0.45
	Calmar	0.35	0.30	0.22	0.25
TDF 90→30 soft term. pain −34%	Wealth	4.08M	4.88M	2.34M	4.14M
	IRR	8.7%	9.3%	6.7%	8.8%
	Max DD (full)	−32.6%	−40.7%	−26.1%	−41.7%
	Final-5yr DD	−22.0%	−22.7%	−17.5%	−23.6%
	Peak-to-trough	1.4y	1.7y	0.3y	1.2y
	Recovery	3.1y	4.5y	1.5y	3.8y
	Sharpe	0.50	0.46	0.41	0.39
	Calmar	0.27	0.23	0.26	0.21
TDF 90→30 hard term. pain −19%	Wealth	2.27M	3.21M	1.98M	3.42M
	IRR	6.6%	7.9%	6.1%	8.1%
	Max DD (full)	−30.0%	−28.2%	−27.9%	−36.6%
	Final-5yr DD	−19.1%	−20.2%	−9.2%	−7.0%
	Peak-to-trough	1.4y	1.4y	0.1y	1.4y
	Recovery	2.4y	4.7y	0.3y	2.7y
	Sharpe	0.42	0.45	0.42	0.44
	Calmar	0.22	0.28	0.22	0.22
TDF 90→3030 soft term. pain −34%	Wealth	4.89M	5.00M	2.33M	1.70M
	IRR	9.4%	9.4%	6.7%	5.5%
	Max DD (full)	−29.9%	−30.2%	−25.0%	−41.3%
	Final-5yr DD	−20.7%	−8.0%	−20.9%	−16.6%
	Peak-to-trough	2.1y	4.6y	0.0y	3.4y
	Recovery	4.2y	7.8y	0.7y	7.4y
	Sharpe	0.53	0.46	0.21	0.18
	Calmar	0.31	0.31	0.27	0.13
TDF 90→3030 hard term. pain −22%	Wealth	3.34M	3.19M	1.91M	1.69M
	IRR	8.0%	7.9%	5.9%	5.4%
	Max DD (full)	−29.0%	−20.8%	−34.5%	−36.4%
	Final-5yr DD	−17.2%	−15.7%	−7.9%	−20.8%
	Peak-to-trough	0.8y	0.4y	0.5y	1.4y
	Recovery	1.5y	1.5y	1.8y	2.8y
	Sharpe	0.48	0.48	0.34	0.27
	Calmar	0.28	0.38	0.17	0.15

1980, tax-free. The figure under each strategy name is its *terminal pain*, the 10th-percentile final-5yr drawdown across all 1,000 paths. Each strategy spans eight rows, one per metric. *Final-5yr DD* is *that path's own* terminal drawdown; *Recovery* is years to recover, or residual DD at the window end.

Appendix G — 1980, tax-free

Strategy	Metric	Actual	MC median by return	MC 10th pct by return	MC 10th pct by max DD
30/30/40 soft term. pain −25%	Wealth	3.64M	3.73M	2.21M	3.87M
	IRR	8.3%	8.4%	6.5%	8.5%
	Max DD (full)	−21.1%	−31.8%	−17.4%	−28.6%
	Final-5yr DD	−17.3%	−31.8%	−12.6%	−28.6%
	Peak-to-trough	0.7y	1.6y	0.8y	0.4y
	Recovery	1.5y	−11% end	2.1y	−21% end
	Sharpe	0.54	0.60	0.45	0.59
	Calmar	0.39	0.26	0.37	0.30
30/30/40 no-reb term. pain −30%	Wealth	4.09M	4.39M	2.32M	3.07M
	IRR	8.7%	9.0%	6.7%	7.7%
	Max DD (full)	−23.8%	−13.8%	−24.9%	−34.7%
	Final-5yr DD	−19.6%	−9.7%	−9.7%	−34.7%
	Peak-to-trough	0.5y	0.1y	0.5y	1.4y
	Recovery	1.4y	0.3y	2.7y	4.2y
	Sharpe	0.56	0.72	0.44	0.49
	Calmar	0.37	0.65	0.27	0.22
30/30/40 hard term. pain −19%	Wealth	3.41M	3.09M	2.04M	3.63M
	IRR	8.1%	7.7%	6.2%	8.3%
	Max DD (full)	−20.4%	−20.1%	−20.2%	−21.8%
	Final-5yr DD	−16.1%	−18.9%	−13.0%	−8.1%
	Peak-to-trough	0.7y	0.7y	0.7y	0.7y
	Recovery	1.4y	1.7y	1.5y	2.1y
	Sharpe	0.57	0.59	0.42	0.69
	Calmar	0.40	0.38	0.31	0.38
40/25/35 soft term. pain −28%	Wealth	4.17M	4.36M	2.42M	4.30M
	IRR	8.8%	9.0%	6.8%	8.9%
	Max DD (full)	−23.4%	−26.6%	−22.4%	−32.0%
	Final-5yr DD	−18.7%	−12.1%	−15.8%	−13.2%
	Peak-to-trough	0.7y	1.1y	0.5y	1.1y
	Recovery	1.5y	2.2y	1.3y	3.9y
	Sharpe	0.57	0.61	0.42	0.70
	Calmar	0.38	0.34	0.31	0.28
40/25/35 no-reb term. pain −33%	Wealth	4.73M	5.12M	2.51M	2.97M
	IRR	9.2%	9.5%	7.0%	7.6%
	Max DD (full)	−28.0%	−25.4%	−24.6%	−39.2%
	Final-5yr DD	−20.8%	−25.4%	−24.6%	−34.8%
	Peak-to-trough	0.5y	0.1y	0.3y	1.7y
	Recovery	1.5y	0.4y	1.2y	4.1y
	Sharpe	0.55	0.55	0.49	0.47
	Calmar	0.33	0.37	0.28	0.19
40/25/35 hard term. pain −22%	Wealth	3.97M	3.72M	2.33M	2.00M
	IRR	8.6%	8.4%	6.7%	6.1%
	Max DD (full)	−24.2%	−15.6%	−15.6%	−24.8%
	Final-5yr DD	−17.3%	−15.6%	−6.4%	−15.6%
	Peak-to-trough	0.5y	0.1y	0.2y	0.5y
	Recovery	1.3y	1.2y	1.1y	1.4y
	Sharpe	0.60	0.64	0.54	0.32
	Calmar	0.36	0.54	0.43	0.25
50/25/25 soft term. pain −31%	Wealth	4.83M	4.98M	2.54M	5.09M
	IRR	9.3%	9.4%	7.0%	9.5%
	Max DD (full)	−27.0%	−23.2%	−26.0%	−36.2%
	Final-5yr DD	−19.5%	−19.4%	−14.3%	−36.2%
	Peak-to-trough	0.7y	0.4y	0.5y	0.5y
	Recovery	1.6y	0.9y	2.6y	3.4y
	Sharpe	0.54	0.56	0.36	0.53
	Calmar	0.34	0.41	0.27	0.26

Appendix G — 1980, tax-free

Strategy	Metric	Actual	MC median by return	MC 10th pct by return	MC 10th pct by max DD
50/25/25 no-reb term. pain −37%	Wealth	5.45M	5.89M	2.67M	9.18M
	IRR	9.7%	10.0%	7.2%	11.5%
	Max DD (full)	−33.1%	−38.0%	−39.6%	−43.3%
	Final-5yr DD	−21.6%	−38.0%	−19.5%	−22.0%
	Peak-to-trough	1.1y	3.7y	2.0y	0.6y
	Recovery	2.4y	4.9y	4.7y	2.2y
	Sharpe	0.52	0.53	0.42	0.55
	Calmar	0.29	0.26	0.18	0.26
50/25/25 hard term. pain −26%	Wealth	4.77M	4.39M	2.47M	4.94M
	IRR	9.3%	9.0%	6.9%	9.4%
	Max DD (full)	−28.9%	−27.7%	−23.6%	−29.9%
	Final-5yr DD	−18.3%	−27.7%	−23.0%	−28.9%
	Peak-to-trough	0.5y	3.8y	0.5y	0.6y
	Recovery	1.4y	4.6y	2.0y	1.6y
	Sharpe	0.57	0.61	0.43	0.67
	Calmar	0.32	0.32	0.29	0.31
40/30/30 soft term. pain −28%	Wealth	4.28M	4.34M	2.35M	3.61M
	IRR	8.9%	8.9%	6.7%	8.3%
	Max DD (full)	−24.7%	−22.7%	−24.0%	−32.7%
	Final-5yr DD	−18.4%	−22.7%	−10.7%	−24.7%
	Peak-to-trough	0.7y	0.2y	0.5y	1.4y
	Recovery	1.6y	−10% end	2.5y	3.9y
	Sharpe	0.53	0.61	0.45	0.54
	Calmar	0.36	0.39	0.28	0.25
40/30/30 no-reb term. pain −33%	Wealth	4.82M	5.11M	2.48M	3.76M
	IRR	9.3%	9.5%	6.9%	8.4%
	Max DD (full)	−28.8%	−29.2%	−20.7%	−39.6%
	Final-5yr DD	−20.6%	−22.5%	−11.2%	−21.2%
	Peak-to-trough	0.5y	0.6y	0.2y	1.2y
	Recovery	1.5y	1.6y	0.9y	2.5y
	Sharpe	0.52	0.53	0.33	0.41
	Calmar	0.32	0.33	0.33	0.21
40/30/30 hard term. pain −23%	Wealth	4.13M	3.67M	2.23M	6.27M
	IRR	8.8%	8.4%	6.5%	10.2%
	Max DD (full)	−25.1%	−13.4%	−23.3%	−26.0%
	Final-5yr DD	−17.0%	−7.4%	−16.7%	−10.4%
	Peak-to-trough	0.5y	0.9y	0.7y	0.7y
	Recovery	1.3y	1.9y	1.6y	2.9y
	Sharpe	0.56	0.78	0.35	0.72
	Calmar	0.35	0.62	0.28	0.39
60/40 soft term. pain −32%	Wealth	5.06M	5.77M	2.90M	5.10M
	IRR	9.5%	9.9%	7.5%	9.5%
	Max DD (full)	−30.7%	−30.4%	−40.2%	−38.6%
	Final-5yr DD	−22.1%	−30.4%	−40.2%	−26.0%
	Peak-to-trough	1.4y	1.4y	1.5y	1.4y
	Recovery	3.0y	3.6y	−30% end	4.3y
	Sharpe	0.61	0.55	0.51	0.53
	Calmar	0.31	0.33	0.19	0.25
60/40 no-reb term. pain −39%	Wealth	5.71M	6.75M	2.91M	5.06M
	IRR	9.9%	10.4%	7.5%	9.5%
	Max DD (full)	−37.4%	−29.2%	−29.3%	−46.4%
	Final-5yr DD	−22.8%	−15.9%	−29.3%	−23.1%
	Peak-to-trough	1.4y	1.1y	3.6y	1.2y
	Recovery	3.2y	2.7y	−21% end	2.5y
	Sharpe	0.56	0.59	0.43	0.45
	Calmar	0.26	0.36	0.26	0.20

Appendix G — 1980, tax-free

Strategy	Metric	Actual	MC median by return	MC 10th pct by return	MC 10th pct by max DD
60/40 hard term. pain −28%	Wealth	4.61M	5.48M	2.99M	1.48M
	IRR	9.2%	9.7%	7.6%	4.9%
	Max DD (full)	−31.7%	−30.7%	−32.0%	−33.4%
	Final-5yr DD	−20.8%	−22.3%	−13.6%	−29.9%
	Peak-to-trough	1.4y	2.0y	1.4y	1.4y
	Recovery	2.4y	3.0y	2.9y	2.7y
	Sharpe	0.62	0.69	0.42	0.31
	Calmar	0.29	0.32	0.24	0.15
PermPort soft term. pain −22%	Wealth	3.25M	3.57M	2.23M	6.46M
	IRR	7.9%	8.3%	6.5%	10.3%
	Max DD (full)	−18.9%	−19.2%	−17.8%	−24.6%
	Final-5yr DD	−18.9%	−19.2%	−11.3%	−24.6%
	Peak-to-trough	0.8y	0.1y	0.8y	0.1y
	Recovery	2.0y	−4% end	2.1y	−8% end
	Sharpe	0.61	0.49	0.45	0.81
	Calmar	0.42	0.43	0.37	0.42
PermPort no-reb term. pain −26%	Wealth	3.64M	4.31M	2.39M	2.71M
	IRR	8.3%	8.9%	6.8%	7.3%
	Max DD (full)	−21.7%	−18.7%	−17.9%	−30.3%
	Final-5yr DD	−21.7%	−11.9%	−15.2%	−22.8%
	Peak-to-trough	0.8y	0.4y	0.6y	0.5y
	Recovery	2.1y	1.9y	2.5y	2.0y
	Sharpe	0.63	0.79	0.56	0.47
	Calmar	0.38	0.48	0.38	0.24
PermPort hard term. pain −16%	Wealth	2.97M	3.00M	2.04M	3.04M
	IRR	7.6%	7.6%	6.2%	7.7%
	Max DD (full)	−17.8%	−17.3%	−16.8%	−18.4%
	Final-5yr DD	−17.8%	−10.1%	−4.3%	−11.7%
	Peak-to-trough	0.9y	0.9y	0.8y	0.9y
	Recovery	2.3y	1.6y	2.3y	1.7y
	Sharpe	0.62	0.53	0.44	0.70
	Calmar	0.43	0.44	0.37	0.42
Conservative soft term. pain −30%	Wealth	3.30M	3.73M	2.07M	2.74M
	IRR	8.0%	8.4%	6.2%	7.3%
	Max DD (full)	−32.0%	−24.5%	−34.9%	−37.7%
	Final-5yr DD	−20.4%	−7.2%	−34.9%	−13.1%
	Peak-to-trough	0.5y	0.1y	1.1y	2.1y
	Recovery	1.5y	0.1y	−12% end	3.8y
	Sharpe	0.50	0.63	0.43	0.45
	Calmar	0.25	0.34	0.18	0.19
Conservative no-reb term. pain −35%	Wealth	3.54M	4.33M	2.21M	3.13M
	IRR	8.2%	8.9%	6.5%	7.8%
	Max DD (full)	−37.2%	−35.3%	−34.2%	−43.6%
	Final-5yr DD	−23.1%	−16.3%	−34.2%	−11.7%
	Peak-to-trough	0.5y	1.7y	0.3y	1.5y
	Recovery	1.9y	3.6y	1.6y	4.1y
	Sharpe	0.48	0.52	0.35	0.50
	Calmar	0.22	0.25	0.19	0.18
Conservative hard term. pain −25%	Wealth	3.35M	3.11M	1.91M	3.63M
	IRR	8.0%	7.8%	5.9%	8.3%
	Max DD (full)	−30.8%	−18.3%	−28.6%	−30.8%
	Final-5yr DD	−18.5%	−6.5%	−12.9%	−30.8%
	Peak-to-trough	0.5y	0.2y	1.1y	0.5y
	Recovery	1.3y	0.4y	2.4y	−8% end
	Sharpe	0.54	0.59	0.47	0.59
	Calmar	0.26	0.42	0.21	0.27

Appendix G — 1980, tax-free

Strategy	Metric	Actual	MC median by return	MC 10th pct by return	MC 10th pct by max DD
Aggressive soft term. pain −55%	Wealth	6.60M	7.07M	2.13M	6.28M
	IRR	10.4%	10.6%	6.3%	10.2%
	Max DD (full)	−63.7%	−44.8%	−44.9%	−63.0%
	Final-5yr DD	−28.5%	−40.2%	−44.9%	−63.0%
	Peak-to-trough	2.5y	0.5y	3.0y	1.6y
	Recovery	6.9y	1.9y	−11% end	4.9y
	Sharpe	0.35	0.55	0.19	0.37
	Calmar	0.16	0.24	0.14	0.16
Aggressive no-reb term. pain −59%	Wealth	8.82M	8.76M	2.29M	12.24M
	IRR	11.3%	11.3%	6.6%	12.4%
	Max DD (full)	−72.7%	−64.4%	−83.3%	−67.4%
	Final-5yr DD	−31.4%	−41.6%	−83.3%	−67.4%
	Peak-to-trough	2.5y	1.4y	9.2y	1.6y
	Recovery	10.5y	6.0y	−74% end	5.0y
	Sharpe	0.37	0.53	0.16	0.44
	Calmar	0.16	0.18	0.08	0.18
Aggressive hard term. pain −40%	Wealth	6.27M	4.39M	1.83M	1.48M
	IRR	10.2%	9.0%	5.8%	4.9%
	Max DD (full)	−40.3%	−37.7%	−55.8%	−47.5%
	Final-5yr DD	−23.8%	−15.8%	−19.7%	−47.5%
	Peak-to-trough	2.6y	0.6y	9.8y	2.8y
	Recovery	4.6y	1.6y	13.0y	−36% end
	Sharpe	0.41	0.34	0.28	0.26
	Calmar	0.25	0.24	0.10	0.10
US/Intl 50/50 soft term. pain −50%	Wealth	4.76M	5.50M	1.98M	1.94M
	IRR	9.3%	9.8%	6.1%	6.0%
	Max DD (full)	−57.4%	−54.7%	−55.8%	−60.2%
	Final-5yr DD	−25.2%	−19.1%	−55.8%	−37.3%
	Peak-to-trough	1.4y	1.0y	1.3y	7.1y
	Recovery	5.1y	3.9y	−37% end	16.1y
	Sharpe	0.38	0.40	0.23	0.25
	Calmar	0.16	0.18	0.11	0.10
US/Intl 50/50 no-reb term. pain −49%	Wealth	5.42M	6.32M	2.30M	7.02M
	IRR	9.7%	10.2%	6.6%	10.6%
	Max DD (full)	−56.4%	−36.9%	−47.1%	−59.3%
	Final-5yr DD	−24.9%	−33.7%	−22.3%	−59.3%
	Peak-to-trough	1.4y	1.5y	0.6y	3.7y
	Recovery	4.9y	3.0y	2.1y	−50% end
	Sharpe	0.40	0.35	0.14	0.44
	Calmar	0.17	0.28	0.14	0.18
US/Intl 50/50 hard term. pain −51%	Wealth	4.38M	4.93M	1.88M	2.27M
	IRR	9.0%	9.4%	5.9%	6.6%
	Max DD (full)	−57.1%	−55.0%	−59.6%	−60.5%
	Final-5yr DD	−25.7%	−55.0%	−50.6%	−60.5%
	Peak-to-trough	1.4y	0.8y	1.2y	8.5y
	Recovery	5.1y	−16% end	2.5y	−18% end
	Sharpe	0.37	0.46	-0.01	0.21
	Calmar	0.16	0.17	0.10	0.11
All Weather soft term. pain −24%	Wealth	3.73M	4.49M	2.65M	2.62M
	IRR	8.4%	9.1%	7.2%	7.1%
	Max DD (full)	−23.3%	−33.4%	−23.8%	−27.3%
	Final-5yr DD	−23.3%	−8.2%	−23.8%	−27.3%
	Peak-to-trough	0.8y	2.0y	2.3y	1.2y
	Recovery	2.3y	5.3y	4.1y	2.4y
	Sharpe	0.69	0.77	0.38	0.47
	Calmar	0.36	0.27	0.30	0.26

Appendix G — 1980, tax-free

Strategy	Metric	Actual	MC median by return	MC 10th pct by return	MC 10th pct by max DD
All Weather no-reb term. pain −27%	Wealth	3.95M	5.08M	2.75M	2.13M
	IRR	8.6%	9.5%	7.3%	6.4%
	Max DD (full)	−24.5%	−24.3%	−19.8%	−32.1%
	Final-5yr DD	−24.5%	−9.9%	−12.2%	−13.1%
	Peak-to-trough	0.8y	0.8y	0.5y	2.4y
	Recovery	2.3y	2.1y	1.3y	4.5y
	Sharpe	0.67	0.66	0.48	0.41
	Calmar	0.35	0.39	0.37	0.20
All Weather hard term. pain −20%	Wealth	3.46M	3.95M	2.49M	6.17M
	IRR	8.1%	8.6%	6.9%	10.1%
	Max DD (full)	−24.0%	−19.9%	−23.9%	−24.0%
	Final-5yr DD	−24.0%	−7.0%	−23.9%	−15.3%
	Peak-to-trough	0.8y	1.1y	0.8y	0.8y
	Recovery	2.6y	2.2y	2.4y	2.1y
	Sharpe	0.69	0.57	0.71	0.99
	Calmar	0.34	0.43	0.29	0.42
Three-fund soft term. pain −32%	Wealth	3.91M	4.59M	2.36M	8.33M
	IRR	8.6%	9.1%	6.7%	11.1%
	Max DD (full)	−34.9%	−43.6%	−30.5%	−39.2%
	Final-5yr DD	−22.3%	−15.2%	−11.6%	−32.1%
	Peak-to-trough	1.4y	1.1y	1.8y	1.0y
	Recovery	3.0y	2.8y	3.2y	3.0y
	Sharpe	0.58	0.63	0.47	0.73
	Calmar	0.25	0.21	0.22	0.28
Three-fund no-reb term. pain −37%	Wealth	4.43M	5.36M	2.46M	13.46M
	IRR	9.0%	9.7%	6.9%	12.7%
	Max DD (full)	−36.8%	−25.3%	−52.1%	−45.0%
	Final-5yr DD	−22.6%	−25.3%	−45.8%	−45.0%
	Peak-to-trough	1.4y	0.4y	1.4y	0.5y
	Recovery	3.0y	2.9y	−19% end	−16% end
	Sharpe	0.55	0.57	0.38	0.71
	Calmar	0.24	0.38	0.13	0.28
Three-fund hard term. pain −29%	Wealth	3.57M	4.23M	2.38M	3.20M
	IRR	8.3%	8.8%	6.8%	7.9%
	Max DD (full)	−33.3%	−20.8%	−34.4%	−34.6%
	Final-5yr DD	−21.2%	−10.6%	−34.4%	−34.6%
	Peak-to-trough	1.4y	0.1y	1.4y	1.3y
	Recovery	2.4y	0.4y	2.9y	3.2y
	Sharpe	0.59	0.60	0.27	0.47
	Calmar	0.25	0.42	0.20	0.23
70/30 soft term. pain −36%	Wealth	5.75M	6.56M	3.06M	2.07M
	IRR	9.9%	10.3%	7.7%	6.2%
	Max DD (full)	−36.4%	−32.5%	−32.8%	−42.2%
	Final-5yr DD	−22.5%	−24.3%	−13.5%	−42.2%
	Peak-to-trough	1.4y	0.8y	0.5y	1.8y
	Recovery	3.1y	1.5y	2.7y	−32% end
	Sharpe	0.58	0.51	0.32	0.37
	Calmar	0.27	0.32	0.24	0.15
70/30 no-reb term. pain −41%	Wealth	6.43M	7.49M	3.07M	14.86M
	IRR	10.3%	10.8%	7.7%	13.0%
	Max DD (full)	−42.3%	−40.9%	−32.8%	−48.8%
	Final-5yr DD	−23.3%	−16.8%	−32.8%	−43.4%
	Peak-to-trough	1.4y	1.0y	3.6y	0.8y
	Recovery	3.3y	3.1y	−25% end	4.4y
	Sharpe	0.53	0.45	0.39	0.61
	Calmar	0.24	0.26	0.24	0.27

Appendix G — 1980, tax-free

Strategy	Metric	Actual	MC median by return	MC 10th pct by return	MC 10th pct by max DD
70/30 hard term. pain −33%	Wealth	5.48M	6.36M	3.16M	5.11M
	IRR	9.7%	10.2%	7.8%	9.5%
	Max DD (full)	−37.7%	−32.7%	−38.0%	−39.3%
	Final-5yr DD	−21.7%	−32.7%	−38.0%	−24.4%
	Peak-to-trough	1.4y	0.5y	0.8y	1.2y
	Recovery	3.0y	1.4y	2.6y	3.7y
	Sharpe	0.57	0.60	0.40	0.50
	Calmar	0.26	0.31	0.21	0.24
80/20 soft term. pain −39%	Wealth	6.50M	7.42M	3.18M	8.33M
	IRR	10.3%	10.8%	7.8%	11.1%
	Max DD (full)	−41.9%	−48.0%	−41.3%	−45.8%
	Final-5yr DD	−23.0%	−41.9%	−26.6%	−45.8%
	Peak-to-trough	1.4y	4.0y	1.0y	1.3y
	Recovery	3.3y	6.3y	2.2y	4.5y
	Sharpe	0.54	0.54	0.22	0.56
	Calmar	0.25	0.22	0.19	0.24
80/20 no-reb term. pain −44%	Wealth	7.16M	8.30M	3.20M	2.81M
	IRR	10.6%	11.1%	7.9%	7.4%
	Max DD (full)	−46.7%	−41.4%	−47.9%	−51.5%
	Final-5yr DD	−23.7%	−32.5%	−17.6%	−51.5%
	Peak-to-trough	1.4y	1.3y	1.2y	1.5y
	Recovery	3.3y	2.7y	3.7y	−16% end
	Sharpe	0.50	0.43	0.37	0.40
	Calmar	0.23	0.27	0.16	0.14
80/20 hard term. pain −38%	Wealth	6.45M	7.44M	3.27M	6.10M
	IRR	10.3%	10.8%	7.9%	10.1%
	Max DD (full)	−43.5%	−42.6%	−33.5%	−45.6%
	Final-5yr DD	−22.5%	−27.7%	−32.7%	−30.9%
	Peak-to-trough	1.4y	1.0y	1.9y	1.7y
	Recovery	3.2y	2.6y	3.4y	4.6y
	Sharpe	0.53	0.49	0.32	0.47
	Calmar	0.24	0.25	0.24	0.22
glA soft term. pain −46%	Wealth	5.15M	6.38M	2.52M	9.27M
	IRR	9.5%	10.3%	7.0%	11.5%
	Max DD (full)	−54.1%	−31.6%	−36.6%	−54.7%
	Final-5yr DD	−27.3%	−31.6%	−35.4%	−15.2%
	Peak-to-trough	2.5y	2.4y	2.1y	3.4y
	Recovery	7.5y	−4% end	3.4y	10.9y
	Sharpe	0.38	0.68	0.35	0.57
	Calmar	0.18	0.32	0.19	0.21
glA hard term. pain −21%	Wealth	3.27M	3.61M	2.27M	3.42M
	IRR	7.9%	8.3%	6.6%	8.1%
	Max DD (full)	−20.5%	−22.0%	−23.5%	−33.4%
	Final-5yr DD	−18.7%	−19.6%	−11.2%	−12.5%
	Peak-to-trough	0.4y	0.4y	0.1y	1.5y
	Recovery	1.8y	0.8y	0.2y	4.1y
	Sharpe	0.42	0.50	0.36	0.27
	Calmar	0.39	0.38	0.28	0.24
glB soft term. pain −48%	Wealth	5.72M	7.12M	2.62M	4.25M
	IRR	9.9%	10.6%	7.1%	8.9%
	Max DD (full)	−55.5%	−28.3%	−39.9%	−56.2%
	Final-5yr DD	−27.3%	−28.3%	−23.6%	−56.2%
	Peak-to-trough	2.5y	0.8y	0.7y	2.9y
	Recovery	7.1y	1.7y	3.4y	−44% end
	Sharpe	0.37	0.67	0.41	0.37
	Calmar	0.18	0.38	0.18	0.16

Appendix G — 1980, tax-free

Strategy	Metric	Actual	MC median by return	MC 10th pct by return	MC 10th pct by max DD
glB hard term. pain −30%	Wealth	4.76M	4.08M	2.24M	3.20M
	IRR	9.3%	8.7%	6.5%	7.9%
	Max DD (full)	−31.8%	−25.6%	−17.5%	−37.3%
	Final-5yr DD	−20.7%	−9.8%	−14.3%	−15.3%
	Peak-to-trough	0.5y	0.9y	0.8y	1.5y
	Recovery	1.5y	1.4y	1.5y	3.8y
	Sharpe	0.48	0.47	0.45	0.36
	Calmar	0.29	0.34	0.37	0.21
glC soft term. pain −34%	Wealth	4.05M	4.89M	2.36M	4.34M
	IRR	8.7%	9.4%	6.7%	8.9%
	Max DD (full)	−34.7%	−44.0%	−47.3%	−48.2%
	Final-5yr DD	−21.7%	−33.1%	−23.0%	−13.1%
	Peak-to-trough	0.2y	1.0y	1.3y	0.9y
	Recovery	1.8y	5.1y	3.0y	3.0y
	Sharpe	0.50	0.45	0.36	0.21
	Calmar	0.25	0.21	0.14	0.19
glC hard term. pain −30%	Wealth	5.27M	4.06M	2.24M	1.88M
	IRR	9.6%	8.7%	6.5%	5.9%
	Max DD (full)	−34.7%	−30.1%	−43.4%	−44.8%
	Final-5yr DD	−20.7%	−9.6%	−23.6%	−44.8%
	Peak-to-trough	0.2y	0.3y	0.5y	1.5y
	Recovery	1.8y	1.6y	1.2y	−25% end
	Sharpe	0.61	0.56	0.12	0.48
	Calmar	0.28	0.29	0.15	0.13
glD soft term. pain −27%	Wealth	4.05M	4.37M	2.34M	5.91M
	IRR	8.7%	9.0%	6.7%	10.0%
	Max DD (full)	−23.0%	−25.0%	−22.7%	−32.9%
	Final-5yr DD	−18.3%	−25.0%	−8.9%	−12.2%
	Peak-to-trough	0.7y	1.0y	0.7y	0.5y
	Recovery	1.5y	−25% end	1.5y	1.9y
	Sharpe	0.54	0.57	0.34	0.64
	Calmar	0.38	0.36	0.30	0.30
glD hard term. pain −24%	Wealth	4.17M	3.83M	2.28M	4.49M
	IRR	8.8%	8.5%	6.6%	9.1%
	Max DD (full)	−21.4%	−25.8%	−17.4%	−27.6%
	Final-5yr DD	−17.1%	−15.6%	−17.4%	−14.5%
	Peak-to-trough	0.7y	0.5y	0.9y	1.3y
	Recovery	1.5y	2.8y	−13% end	4.0y
	Sharpe	0.58	0.50	0.51	0.58
	Calmar	0.41	0.33	0.38	0.33
GEM monthly term. pain −34%	Wealth	9.54M	7.00M	3.09M	3.25M
	IRR	11.6%	10.6%	7.7%	7.9%
	Max DD (full)	−32.4%	−28.0%	−24.2%	−40.8%
	Final-5yr DD	−29.4%	−27.1%	−16.9%	−27.2%
	Peak-to-trough	0.1y	1.5y	0.7y	1.1y
	Recovery	1.5y	3.2y	3.6y	2.6y
	Sharpe	0.67	0.62	0.38	0.26
	Calmar	0.36	0.38	0.32	0.19
GEM quarterly term. pain −35%	Wealth	6.40M	6.50M	2.84M	2.22M
	IRR	10.3%	10.3%	7.4%	6.5%
	Max DD (full)	−47.5%	−34.1%	−48.3%	−49.8%
	Final-5yr DD	−26.3%	−34.1%	−20.4%	−19.8%
	Peak-to-trough	0.3y	0.3y	0.3y	1.7y
	Recovery	3.5y	2.2y	5.0y	9.2y
	Sharpe	0.51	0.54	0.19	0.27
	Calmar	0.22	0.30	0.15	0.13

Appendix G — 1980, tax-free

Strategy	Metric	Actual	MC median by return	MC 10th pct by return	MC 10th pct by max DD
TrendFollow term. pain −29%	Wealth	4.05M	4.18M	2.26M	5.62M
	IRR	8.7%	8.8%	6.6%	9.8%
	Max DD (full)	−23.9%	−23.8%	−23.0%	−33.1%
	Final-5yr DD	−18.1%	−10.6%	−10.6%	−6.8%
	Peak-to-trough	0.7y	2.7y	0.7y	0.3y
	Recovery	1.5y	3.5y	2.7y	1.6y
	Sharpe	0.51	0.50	0.37	0.56
	Calmar	0.36	0.37	0.29	0.30
MeanRevert term. pain −27%	Wealth	3.92M	3.95M	2.22M	9.47M
	IRR	8.6%	8.6%	6.5%	11.6%
	Max DD (full)	−23.1%	−21.1%	−29.5%	−31.0%
	Final-5yr DD	−17.5%	−21.1%	−5.5%	−19.9%
	Peak-to-trough	0.7y	0.2y	1.9y	2.1y
	Recovery	1.5y	−9% end	7.4y	4.1y
	Sharpe	0.51	0.62	0.37	0.82
	Calmar	0.37	0.41	0.22	0.37
TF+MR term. pain −28%	Wealth	3.96M	4.06M	2.24M	3.39M
	IRR	8.6%	8.7%	6.5%	8.1%
	Max DD (full)	−23.3%	−25.9%	−40.0%	−31.8%
	Final-5yr DD	−17.7%	−21.4%	−8.6%	−24.2%
	Peak-to-trough	0.7y	0.2y	2.1y	1.4y
	Recovery	1.5y	1.0y	8.9y	2.8y
	Sharpe	0.52	0.59	0.54	0.53
	Calmar	0.37	0.34	0.16	0.25
TDF 90→30 soft term. pain −33%	Wealth	5.05M	6.21M	2.93M	5.55M
	IRR	9.5%	10.2%	7.5%	9.8%
	Max DD (full)	−31.9%	−37.1%	−34.3%	−40.9%
	Final-5yr DD	−21.9%	−28.4%	−22.5%	−40.9%
	Peak-to-trough	1.4y	1.1y	0.8y	2.5y
	Recovery	3.0y	3.6y	1.4y	5.5y
	Sharpe	0.57	0.50	0.33	0.33
	Calmar	0.30	0.27	0.22	0.24
TDF 90→30 hard term. pain −18%	Wealth	3.34M	4.93M	2.85M	6.16M
	IRR	8.0%	9.4%	7.4%	10.1%
	Max DD (full)	−28.8%	−37.1%	−35.7%	−35.9%
	Final-5yr DD	−18.5%	−13.1%	−6.3%	−7.3%
	Peak-to-trough	1.4y	1.3y	3.0y	0.8y
	Recovery	2.2y	2.6y	4.4y	1.6y
	Sharpe	0.54	0.51	0.48	0.71
	Calmar	0.28	0.25	0.21	0.28
TDF 90→3030 soft term. pain −34%	Wealth	5.74M	6.17M	2.85M	5.74M
	IRR	9.9%	10.1%	7.4%	9.9%
	Max DD (full)	−30.1%	−38.8%	−24.7%	−41.0%
	Final-5yr DD	−20.8%	−24.4%	−18.3%	−41.0%
	Peak-to-trough	0.5y	3.1y	0.3y	0.7y
	Recovery	1.5y	5.7y	1.5y	−10% end
	Sharpe	0.58	0.39	0.46	0.47
	Calmar	0.33	0.26	0.30	0.24
TDF 90→3030 hard term. pain −20%	Wealth	4.73M	4.65M	2.69M	2.60M
	IRR	9.2%	9.2%	7.2%	7.1%
	Max DD (full)	−28.6%	−33.5%	−29.3%	−35.7%
	Final-5yr DD	−16.8%	−30.2%	−5.5%	−8.1%
	Peak-to-trough	0.5y	1.1y	0.5y	0.8y
	Recovery	1.5y	2.7y	4.0y	2.3y
	Sharpe	0.59	0.50	0.42	0.32
	Calmar	0.32	0.27	0.25	0.20